Chemical Mixtures and Accelerated Biological Aging in CKD: NHANES Analysis

Reproducible NHANES analysis By January

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Study Objective

Primary question: are environmental chemical mixtures associated with accelerated biological aging (PhenoAge acceleration), and is this association stronger among individuals with CKD?

Configuration

```r
config <- list(
  cycles = c("2003-2004", "2005-2006", "2007-2008", "2009-2010", "2011-2012", "2013-2014", "2015-2016",
  exposure_family = "pfas", # pfas, metals, both
  q = 4,
  min_age = 20,
  ckd_def = "egfr_or_acr", # egfr_or_acr, egfr_only, acr_only
  use_residualized_phenoage = TRUE,   # PhenoAgeAccel is always residualized; option reserved for futur
  # Primary analysis uses fixed 4-PFAS set with complete-case; strict (qgcomp-compatible subset) is sen
  require_complete_exposure_set = FALSE,  # Legacy: primary is always fixed-set complete-case; no longe
  min_nonmissing_exposure_fraction = 0.6,
  min_nonmissing_exposures_floor = 2,
  include_clinical_covariates = TRUE,    # Not wired to primary (6-variable set); reserved for future s
  # Weight is resolved from audit; PFAS workflow prioritizes environmental subsample WTS* weights.
  weight_type = "auto",  # Auto-resolve from audit in the exposure-complete sample (PFAS excludes WTINT
  verbose = TRUE
)
config
```

```
## $cycles
## [1] "2003-2004" "2005-2006" "2007-2008" "2009-2010" "2011-2012" "2013-2014"
## [7] "2015-2016" "2017-2018"
##
## $exposure_family
## [1] "pfas"
##
## $q
## [1] 4
##
## $min_age
## [1] 20
##
## $ckd_def
## [1] "egfr_or_acr"
##
## $use_residualized_phenoage
## [1] TRUE
##
## $require_complete_exposure_set
## [1] FALSE
##
## $min_nonmissing_exposure_fraction
## [1] 0.6
##
## $min_nonmissing_exposures_floor
## [1] 2
##
## $include_clinical_covariates
## [1] TRUE
##
## $weight_type
```

```
## [1] "auto"
##
## $verbose
## [1] TRUE
```

Packages

```r
required_pkgs <- c(
  "nhanesA", "dplyr", "tidyr", "purrr", "stringr", "survey", "splines",
  "ggplot2", "broom", "knitr", "qgcomp", "forcats", "tibble"
)
invisible(lapply(required_pkgs, function(p) {
  if (!requireNamespace(p, quietly = TRUE)) stop("Package not installed: ", p)
}))

library(nhanesA)
library(dplyr)
library(tidyr)
library(purrr)
library(stringr)
library(survey)
options(survey.lonely.psu = "adjust")
library(splines)
library(ggplot2)
library(broom)
library(knitr)
library(qgcomp)
library(forcats)
library(tibble)
```

Helper Functions

```r
cycle_to_suffix <- function(cycle) {
  map <- c(
    "1999-2000" = "A", "2001-2002" = "B", "2003-2004" = "C", "2005-2006" = "D",
    "2007-2008" = "E", "2009-2010" = "F", "2011-2012" = "G", "2013-2014" = "H",
    "2015-2016" = "I", "2017-2018" = "J", "2019-2020" = "K"
  )
  out <- unname(map[cycle])
  if (is.na(out)) stop("Cycle not mapped: ", cycle)
  out
}

safe_nhanes <- function(table_name, verbose = TRUE) {
  out <- tryCatch(nhanesA::nhanes(table_name), error = function(e) NULL)
  if (is.null(out) && verbose) message("Unavailable table: ", table_name)
  out
}
```

```r
resolve_table <- function(cycle, candidates, domain = NULL, verbose = TRUE) {
  suf <- cycle_to_suffix(cycle)
  to_try <- paste0(candidates, "_", suf)
  for (tb in to_try) {
    dat <- safe_nhanes(tb, verbose = FALSE)
    if (!is.null(dat)) {
      if (verbose) message("Using table ", tb, " for cycle ", cycle)
      return(list(code = candidates[1], table = tb, data = dat))
    }
  }
  NULL
}

coalesce_num <- function(df, vars) {
  vars <- vars[vars %in% names(df)]
  if (length(vars) == 0) return(rep(NA_real_, nrow(df)))
  v <- lapply(df[vars], function(x) suppressWarnings(as.numeric(as.character(x))))
  dplyr::coalesce(!!!v)
}

parse_yes_no <- function(x) {
  x_num <- suppressWarnings(as.numeric(as.character(x)))
  x_chr <- toupper(trimws(as.character(x)))
  case_when(
    x_num == 1 ~ 1,
    x_num == 2 ~ 0,
    str_detect(x_chr, "^YES$") ~ 1,
    str_detect(x_chr, "^NO$") ~ 0,
    TRUE ~ NA_real_
  )
}

parse_smq040 <- function(x) {
  x_chr <- toupper(trimws(as.character(x)))
  x_num <- suppressWarnings(as.numeric(as.character(x)))
  case_when(
    x_num %in% c(1, 2) ~ 1,
    x_num == 3 ~ 0,
    str_detect(x_chr, "EVERY DAY|SOME DAYS") ~ 1,
    str_detect(x_chr, "^NOT AT ALL$") ~ 0,
    TRUE ~ NA_real_
  )
}

parse_phq_item <- function(x) {
  x_chr <- toupper(trimws(as.character(x)))
  x_num <- suppressWarnings(as.numeric(as.character(x)))
  case_when(
    x_num == 1 ~ 0, x_num == 2 ~ 1, x_num == 3 ~ 2, x_num == 4 ~ 3,
    x_num %in% 0:3 ~ x_num,
    str_detect(x_chr, "^NOT AT ALL$") ~ 0,
    str_detect(x_chr, "^SEVERAL DAYS$") ~ 1,
    str_detect(x_chr, "^MORE THAN HALF THE DAYS$") ~ 2,
```

```r
    str_detect(x_chr, "^NEARLY EVERY DAY$") ~ 3,
    TRUE ~ NA_real_
  )
}

yesno01 <- function(x) {
  xc <- toupper(trimws(as.character(x)))
  xn <- suppressWarnings(as.numeric(as.character(x)))
  case_when(
    xn == 1 ~ 1,
    xn == 2 ~ 0,
    xc %in% c("YES", "1") ~ 1,
    xc %in% c("NO", "2") ~ 0,
    TRUE ~ NA_real_
  )
}

compute_egfr_2021 <- function(scr_mgdl, age, sex_raw) {
  sex_chr <- toupper(trimws(as.character(sex_raw)))
  sex_num <- suppressWarnings(as.numeric(as.character(sex_raw)))
  male <- case_when(
    sex_num == 1 ~ TRUE,
    sex_num == 2 ~ FALSE,
    sex_chr == "MALE" ~ TRUE,
    sex_chr == "FEMALE" ~ FALSE,
    TRUE ~ NA
  )
  kappa <- ifelse(male, 0.9, 0.7)
  alpha <- ifelse(male, -0.302, -0.241)
  sex_mult <- ifelse(male, 1, 1.012)
  142 * pmin(scr_mgdl / kappa, 1)^alpha * pmax(scr_mgdl / kappa, 1)^(-1.200) * (0.9938^age) * sex_mult
}

compute_acr <- function(ua, ucr, acr_direct = NULL) {
  calc <- ifelse(!is.na(ua) & !is.na(ucr) & ucr > 0, (ua / ucr) * 100, NA_real_)
  dplyr::coalesce(acr_direct, calc)
}

compute_phenoage_components <- function(df, include_scr = TRUE, include_alb = TRUE) {
  # Levine et al. (2018) coefficients expect: albumin g/L, creatinine µmol/L, glucose mmol/L.
  # NHANES provides: albumin g/dL, creatinine mg/dL, glucose often mg/dL (we standardize to mg/dL in pi
  alb_gL   <- df$alb * 10
  scr_umol <- df$scr * 88.4
  glu_mmol <- df$glu / 18.018
  alb_term <- if (include_alb) (-0.0336 * alb_gL) else 0
  scr_term <- if (include_scr) ( 0.0095 * scr_umol) else 0
  glu_term <- 0.1953 * glu_mmol

  xb <- -19.907 +
    alb_term +
    scr_term +
    glu_term +
    0.0954 * log(pmax(df$crp, 1e-6)) -
```

```r
    0.0120 * df$lym +
    0.0268 * df$mcv +
    0.3306 * df$rdw +
    0.00188 * df$alp +
    0.0554 * df$wbc +
    0.0804 * df$age

  xb <- pmin(pmax(xb, -50), 50)
  gamma_const <- (exp(120 * 0.0076927) - 1) / 0.0076927
  hazard <- exp(xb) * gamma_const
  mort <- -expm1(-hazard)
  mort <- pmin(pmax(mort, 1e-12), 1 - 1e-12)
  pheno <- 141.50 + log(-0.00553 * log1p(-mort)) / 0.090165

  list(xb = xb, mort = mort, phenoage = pheno)
}

safe_phenoage <- function(df) {
  compute_phenoage_components(df, include_scr = TRUE, include_alb = TRUE)$phenoage
}

safe_qgcomp_extract <- function(fit_obj, model_label, n_obs) {
  if (is.null(fit_obj)) {
    return(tibble(model = model_label, beta = NA_real_, conf.low = NA_real_, conf.high = NA_real_, p.val
  }
  cf <- coef(fit_obj)
  psi_name <- grep("^psi", names(cf), value = TRUE)[1]
  if (is.na(psi_name)) psi_name <- grep("psi", names(cf), value = TRUE)[1]
  if (is.na(psi_name)) {
    message("qgcomp terms for [", model_label, "]: no psi found in ", paste(names(cf), collapse = ", "))
    return(tibble(model = model_label, beta = NA_real_, conf.low = NA_real_, conf.high = NA_real_, p.val
  }
  beta <- unname(cf[psi_name])
  vc <- tryCatch(vcov(fit_obj), error = function(e) NULL)
  if (!is.null(vc) && psi_name %in% rownames(vc) && psi_name %in% colnames(vc)) {
    se <- sqrt(vc[psi_name, psi_name])
    lo <- beta - 1.96 * se
    hi <- beta + 1.96 * se
    p_val <- 2 * pnorm(-abs(beta / se))
  } else {
    lo <- hi <- p_val <- NA_real_
  }
  tibble(model = model_label, beta = beta, conf.low = lo, conf.high = hi, p.value = p_val, N = n_obs)
}

resolve_weight_var <- function(df, weight_type = "auto", exposure_family = "pfas") {
  cols <- names(df)
  if (weight_type != "auto" && weight_type != "MEC") {
    if (!(weight_type %in% cols)) stop("Requested weight variable not found: ", weight_type)
    return(list(var = weight_type, note = paste("User-specified weight:", weight_type)))
  }
  if ("WTMEC2YR" %in% cols) return(list(var = "WTMEC2YR", note = "Generic fallback: using MEC weight WT
  stop("No supported NHANES weight variable found.")
```

```r
}

apply_lod_rule <- function(df, varname) {
  x <- suppressWarnings(as.numeric(as.character(df[[varname]])))

  base <- sub("^(LBX|URX)", "", varname)
  lc_candidates <- c(
    paste0(varname, "LC"),
    paste0("LBD", base, "LC"),
    paste0("URD", base, "LC"),
    # NHANES often uses *L (not *LC) as the detection-limit flag.
    paste0(varname, "L"),
    paste0("LBD", base, "L"),
    paste0("URD", base, "L")
  )
  lc_candidates <- lc_candidates[lc_candidates %in% names(df)]

  if (length(lc_candidates) > 0) {
    lc <- df[[lc_candidates[1]]]
    lc_num <- suppressWarnings(as.numeric(as.character(lc)))
    lc_chr <- toupper(trimws(as.character(lc)))
    below <- (!is.na(lc_num) & lc_num == 1) | str_detect(lc_chr, "BELOW|<")
    below[is.na(below)] <- FALSE
    # NHANES convention: when LC==1, the concentration column typically holds the LOD value.
    # Standard replacement: value / sqrt(2).
    x[below] <- x[below] / sqrt(2)
    # If value is still NA/non-positive after that, use smallest observed LOD / sqrt(2).
    lod_vals <- x[below & x > 0]
    fallback <- if (length(lod_vals) > 0) min(lod_vals, na.rm = TRUE) else NA_real_
    x[below & (is.na(x) | x <= 0)] <- fallback
    return(x)
  }
  # If no LOD indicator exists, do not impute missing values.
  x
}
```

Data Acquisition and Cycle Availability

```r
# Cycle-aware: 2003-2004 (C) uses Lxx_ names (L40, L25, L06CRP, L06ALB, L24PFC, L06BMT, L06UHM); 2005+
table_candidates <- list(
  DEMO = c("DEMO"),
  BMX = c("BMX"),
  SMQ = c("SMQ"),
  ALQ = c("ALQ"),
  PAQ = c("PAQ"),
  DIQ = c("DIQ"),
  BPQ = c("BPQ"),
  MCQ = c("MCQ"),
  BIOPRO = c("L40", "BIOPRO"),
  CBC = c("L25", "CBC"),
  CRP = c("L06CRP", "CRP", "HSCRP"),
```

```r
  ALB_CR = c("L06ALB", "ALB_CR"),
  GLU = c("L10", "GLU"),
  PFAS = c("L24PFC", "PFAS", "PFC", "P_PFAS"),
  BLOOD_METALS = c("L06BMT", "PBCD"),
  URINE_METALS = c("L06UHM", "UHM", "UM")
)

download_cycle <- function(cycle, exposure_family = "pfas", verbose = TRUE) {
  core_domains <- c("DEMO", "BMX", "SMQ", "ALQ", "PAQ", "DIQ", "BPQ", "MCQ", "BIOPRO", "CBC", "CRP", "A
  exp_domains <- switch(
    exposure_family,
    pfas = c("PFAS"),
    metals = c("BLOOD_METALS", "URINE_METALS"),
    both = c("PFAS", "BLOOD_METALS", "URINE_METALS"),
    c("PFAS")
  )
  all_domains <- c(core_domains, exp_domains)

  resolved <- lapply(all_domains, function(dom) resolve_table(cycle, table_candidates[[dom]], domain = 
  names(resolved) <- all_domains
  resolved <- resolved[!vapply(resolved, is.null, logical(1))]
  if (!"DEMO" %in% names(resolved)) return(NULL)

  dat <- resolved$DEMO$data
  if (!"SEQN" %in% names(dat)) return(NULL)

  for (nm in setdiff(names(resolved), "DEMO")) {
    d <- resolved[[nm]]$data
    # Rename glucose vars from GLU table before joining so they do not collide with BIOPRO/L40.
    if (nm == "GLU") {
      if ("LBXGLU" %in% names(d)) names(d)[names(d) == "LBXGLU"] <- "LBXGLU_GLU"
      if ("LBXSGL" %in% names(d)) names(d)[names(d) == "LBXSGL"] <- "LBXSGL_GLU"
    }
    if ("SEQN" %in% names(d)) dat <- left_join(dat, d, by = "SEQN")
  }
  # After merge: define explicit glucose sources. Primary analysis uses MEC chemistry glucose for measu
  dat$glu_mec  <- coalesce_num(dat, c("LBXGLU", "LBXSGL"))
  dat$glu_glu  <- coalesce_num(dat, c("LBXGLU_GLU", "LBXSGL_GLU"))
  dat$glu     <- dat$glu_mec
  dat$cycle <- cycle
  # Store provenance for key tables so we can audit weight/source compatibility.
  if ("CRP" %in% names(resolved)) dat$crp_table <- resolved$CRP$table
  if ("GLU" %in% names(resolved)) dat$glu_table <- resolved$GLU$table
  # Warn if joins created .x/.y duplicate columns (would break coalesce_num).
  dup_suffix_cols <- names(dat)[grepl("\\.x$|\\.y$", names(dat))]
  if (length(dup_suffix_cols) > 0) {
    warning("Cycle ", cycle, ": duplicate columns from join: ", paste(dup_suffix_cols, collapse = ", ")
  }
  dat
}

cycle_data <- lapply(config$cycles, download_cycle, exposure_family = config$exposure_family, verbose =
cycle_data <- cycle_data[!vapply(cycle_data, is.null, logical(1))]
```

```r
if (length(cycle_data) == 0) stop("No cycles were successfully downloaded.")

all_cols <- sort(unique(unlist(lapply(cycle_data, names))))
cycle_data <- lapply(cycle_data, function(df) {
  miss <- setdiff(all_cols, names(df))
  if (length(miss) > 0) for (m in miss) df[[m]] <- NA
  df <- df[, all_cols, drop = FALSE]
  df %>% mutate(across(everything(), as.character))
})
nhanes_raw <- bind_rows(cycle_data)

kable(
  data.frame(
    cycle = names(table(config$cycles))[names(table(config$cycles)) %in% unique(nhanes_raw$cycle)],
    n = as.integer(table(nhanes_raw$cycle))
  ),
  caption = "Cycle availability and downloaded participant counts"
)
```

Table 1: Cycle availability and downloaded participant counts

cycle	n
2003-2004	10122
2005-2006	10348
2007-2008	10149
2009-2010	10537
2011-2012	9756
2013-2014	10175
2015-2016	9971
2017-2018	9254

Variable Construction: PhenoAge, CKD, Covariates

```r
dpq_vars <- names(nhanes_raw)[str_detect(names(nhanes_raw), "^DPQ0")]

dat <- nhanes_raw %>%
  mutate(across(any_of(dpq_vars), parse_phq_item)) %>%
  mutate(
    age = suppressWarnings(as.numeric(as.character(RIDAGEYR))),
    sex = case_when(
      suppressWarnings(as.numeric(as.character(RIAGENDR))) == 1 ~ "Male",
      suppressWarnings(as.numeric(as.character(RIAGENDR))) == 2 ~ "Female",
      toupper(trimws(as.character(RIAGENDR))) == "MALE" ~ "Male",
      toupper(trimws(as.character(RIAGENDR))) == "FEMALE" ~ "Female",
      TRUE ~ NA_character_
    ),
    race_eth = {
      r3 <- suppressWarnings(as.numeric(as.character(RIDRETH3)))
      r1 <- suppressWarnings(as.numeric(as.character(RIDRETH1)))
```

```r
      r3_chr <- toupper(trimws(as.character(RIDRETH3)))
      r1_chr <- toupper(trimws(as.character(RIDRETH1)))
      case_when(
        !is.na(r3) & r3 %in% c(1, 2) ~ "Hispanic",
        !is.na(r3) & r3 == 3 ~ "Non-Hispanic White",
        !is.na(r3) & r3 == 4 ~ "Non-Hispanic Black",
        !is.na(r3) & r3 == 6 ~ "Non-Hispanic Asian",
        !is.na(r3) & r3 %in% c(5, 7) ~ "Other/Multi",
        is.na(r3) & r1 %in% c(1, 2) ~ "Hispanic",
        is.na(r3) & r1 == 3 ~ "Non-Hispanic White",
        is.na(r3) & r1 == 4 ~ "Non-Hispanic Black",
        is.na(r3) & r1 == 5 ~ "Other/Multi",
        str_detect(r3_chr, "NON-HISPANIC WHITE") ~ "Non-Hispanic White",
        str_detect(r3_chr, "NON-HISPANIC BLACK") ~ "Non-Hispanic Black",
        str_detect(r3_chr, "NON-HISPANIC ASIAN") ~ "Non-Hispanic Asian",
        str_detect(r3_chr, "OTHER RACE|MULTI") ~ "Other/Multi",
        str_detect(r3_chr, "^MEXICAN AMERICAN$|^OTHER HISPANIC$|^HISPANIC$") ~ "Hispanic",
        str_detect(r1_chr, "NON-HISPANIC WHITE") ~ "Non-Hispanic White",
        str_detect(r1_chr, "NON-HISPANIC BLACK") ~ "Non-Hispanic Black",
        str_detect(r1_chr, "OTHER RACE|MULTI") ~ "Other/Multi",
        str_detect(r1_chr, "^MEXICAN AMERICAN$|^OTHER HISPANIC$|^HISPANIC$") ~ "Hispanic",
        TRUE ~ NA_character_
      )
    },
    education = {
      e_num <- suppressWarnings(as.numeric(as.character(DMDEDUC2)))
      e_chr <- toupper(trimws(as.character(DMDEDUC2)))
      case_when(
        e_num %in% c(1, 2) ~ "<HS",
        e_num == 3 ~ "HS/GED",
        e_num %in% c(4, 5) ~ ">HS",
        str_detect(e_chr, "LESS THAN 9TH|9TH.*12TH") ~ "<HS",
        str_detect(e_chr, "HIGH SCHOOL|GED") ~ "HS/GED",
        str_detect(e_chr, "SOME COLLEGE|COLLEGE GRADUATE") ~ ">HS",
        TRUE ~ NA_character_
      )
    },
    pir = suppressWarnings(as.numeric(as.character(INDFMPIR))),
    bmi = suppressWarnings(as.numeric(as.character(BMXBMI))),
    smoker = {
      smq040_chr <- toupper(trimws(as.character(SMQ040)))
      smq040_num <- suppressWarnings(as.numeric(as.character(SMQ040)))
      case_when(
        smq040_num %in% c(1, 2) ~ 1,
        smq040_num == 3 ~ 0,
        str_detect(smq040_chr, "EVERY DAY|SOME DAYS") ~ 1,
        str_detect(smq040_chr, "^NOT AT ALL$") ~ 0,
        TRUE ~ NA_real_
      )
    },
    alcohol = parse_yes_no(ALQ151),
    phys_active = case_when(
      parse_yes_no(PAQ650) == 1 | parse_yes_no(PAQ665) == 1 ~ 1,
```

```r
      parse_yes_no(PAQ650) == 0 & parse_yes_no(PAQ665) == 0 ~ 0,
      TRUE ~ NA_real_
    ),
    diabetes = yesno01(DIQ010),
    hypertension = yesno01(BPQ020),
    cvd = case_when(
      parse_yes_no(MCQ160B) == 1 | parse_yes_no(MCQ160C) == 1 | parse_yes_no(MCQ160E) == 1 | parse_yes_
      parse_yes_no(MCQ160B) == 0 & parse_yes_no(MCQ160C) == 0 & parse_yes_no(MCQ160E) == 0 & parse_yes_
      TRUE ~ NA_real_
    ),
    phenoage_accel = NA_real_
  )

# Compute biomarkers, CKD markers, and design variables outside mutate (avoids cur_data_all()).
dat$alb <- coalesce_num(dat, c("LBXSAL"))
dat$scr <- coalesce_num(dat, c("LBXSCR", "LBDSCRSI"))
# Glucose: glu_mec / glu_glu / glu set in download_cycle (primary uses glu_mec by design).
if (!"glu" %in% names(dat)) dat$glu <- dat$glu_mec
dat$crp <- coalesce_num(dat, c("LBXCRP", "LBXHSCRP"))
dat$lym <- coalesce_num(dat, c("LBXLYPCT"))
dat$mcv <- coalesce_num(dat, c("LBXMCV", "LBXMCVSI"))
dat$rdw <- coalesce_num(dat, c("LBXRW", "LBXRDW"))
dat$alp <- coalesce_num(dat, c("LBXAPSI", "LBXSAPSI"))
dat$wbc <- coalesce_num(dat, c("LBXWBCSI", "LBXWBC"))

dat$ualb <- coalesce_num(dat, c("URXUMA"))
dat$ucr <- coalesce_num(dat, c("URXUCR", "URDUCR"))
dat$acr_direct <- coalesce_num(dat, c("URDACT"))
dat$acr <- compute_acr(dat$ualb, dat$ucr, dat$acr_direct)
dat$egfr <- compute_egfr_2021(dat$scr, dat$age, dat$RIAGENDR)
dat$ckd_egfr <- ifelse(!is.na(dat$egfr), as.integer(dat$egfr < 60), NA_integer_)
dat$ckd_acr <- ifelse(!is.na(dat$acr), as.integer(dat$acr >= 30), NA_integer_)
dat$ckd <- case_when(
  config$ckd_def == "egfr_or_acr" & (dat$ckd_egfr == 1 | dat$ckd_acr == 1) ~ 1L,
  config$ckd_def == "egfr_or_acr" & (!is.na(dat$ckd_egfr) | !is.na(dat$ckd_acr)) ~ 0L,
  config$ckd_def == "egfr_only" ~ dat$ckd_egfr,
  config$ckd_def == "acr_only" ~ dat$ckd_acr,
  TRUE ~ NA_integer_
)
dat$ckd_stage <- cut(
  dat$egfr, breaks = c(-Inf, 15, 30, 45, 60, 90, Inf),
  labels = c("G5", "G4", "G3b", "G3a", "G2", "G1"),
  right = FALSE  # KDIGO: left-closed intervals [lo, hi). E.g., 60 -> G2, 59.9 -> G3a.
)
dat$strata <- coalesce_num(dat, c("SDMVSTRA"))
dat$psu <- coalesce_num(dat, c("SDMVPSU"))

# Coding audits (raw -> derived) to prevent silent recode failures.
kable(
  dat %>% count(cycle, ridreth3_missing = is.na(RIDRETH3)) %>% arrange(cycle, ridreth3_missing),
  caption = "Race audit: availability of RIDRETH3 by cycle"
)
```

Table 2: Race audit: availability of RIDRETH3 by cycle

cycle	ridreth3_missing	n
2003-2004	TRUE	10122
2005-2006	TRUE	10348
2007-2008	TRUE	10149
2009-2010	TRUE	10537
2011-2012	FALSE	9756
2013-2014	FALSE	10175
2015-2016	FALSE	9971
2017-2018	FALSE	9254

```
kable(as.data.frame(table(dat$RIDRETH1, useNA = "ifany")), caption = "Raw RIDRETH1 distribution")
```

Table 3: Raw RIDRETH1 distribution

Var1	Freq
Mexican American	16280
Non-Hispanic Black	18735
Non-Hispanic White	29459
Other Hispanic	7188
Other Race - Including Multi-Racial	8650

```
kable(as.data.frame(table(dat$RIDRETH3, useNA = "ifany")), caption = "Raw RIDRETH3 distribution")
```

Table 4: Raw RIDRETH3 distribution

Var1	Freq
Mexican American	6373
Non-Hispanic Asian	4566
Non-Hispanic Black	9194
Non-Hispanic White	12863
Other Hispanic	4164
Other Race - Including Multi-Racial	1996
NA	41156

```
kable(as.data.frame(table(dat$race_eth, useNA = "ifany")), caption = "Derived race_eth distribution")
```

Table 5: Derived race_eth distribution

Var1	Freq
Hispanic	23468
Non-Hispanic Asian	4566
Non-Hispanic Black	18735
Non-Hispanic White	29459
Other/Multi	4084

```r
kable(as.data.frame(table(dat$SMQ040, useNA = "ifany")), caption = "Raw SMQ040 distribution")
```

Table 6: Raw SMQ040 distribution

Var1	Freq
Every day	7593
Not at all	10892
Refused	3
Some days	1753
NA	60071

```r
kable(as.data.frame(table(dat$DIQ010, useNA = "ifany")), caption = "Raw DIQ010 distribution")
```

Table 7: Raw DIQ010 distribution

Var1	Freq
Borderline	1062
Don't know	48
No	69943
Refused	3
Yes	5790
NA	3466

```r
kable(as.data.frame(table(dat$BPQ020, useNA = "ifany")), caption = "Raw BPQ020 distribution")
```

Table 8: Raw BPQ020 distribution

Var1	Freq
Don't know	83
No	34534
Yes	16221
NA	29474

```r
kable(as.data.frame(table(dat$smoker, useNA = "ifany")), caption = "Derived smoker distribution")
```

Table 9: Derived smoker distribution

Var1	Freq
0	10892
1	9346
NA	60074

```r
kable(as.data.frame(table(dat$diabetes, useNA = "ifany")), caption = "Derived diabetes distribution")
```

Table 10: Derived diabetes distribution

Var1	Freq
0	69943
1	5790
NA	4579

```
kable(as.data.frame(table(dat$hypertension, useNA = "ifany")), caption = "Derived hypertension distribu
```

Table 11: Derived hypertension distribution

Var1	Freq
0	34534
1	16221
NA	29557

```
# Codebook-aligned harmonization checks (raw coding support by cycle).
check_raw_levels <- function(df, var, allowed_patterns) {
  if (!(var %in% names(df))) return(tibble(variable = var, cycle = NA_character_, n_unexpected = NA_int
  df %>%
    mutate(.raw = toupper(trimws(as.character(.data[[var]])))) %>%
    group_by(cycle) %>%
    summarise(
      n_nonmissing = sum(!is.na(.raw) & .raw != ""),
      n_unexpected = sum(!is.na(.raw) & .raw != "" & !str_detect(.raw, allowed_patterns)),
      unexpected_examples = paste(head(unique(.raw[!is.na(.raw) & .raw != "" & !str_detect(.raw, allowe
      .groups = "drop"
    ) %>%
    mutate(variable = var) %>%
    select(variable, cycle, n_nonmissing, n_unexpected, unexpected_examples)
}

harmonization_checks <- bind_rows(
  check_raw_levels(dat, "RIAGENDR", "^(1|2|MALE|FEMALE)$"),
  check_raw_levels(dat, "DMDEDUC2", "^(1|2|3|4|5|7|9)$|LESS THAN 9TH|9-11TH|HIGH SCHOOL|GED|SOME COLLEG
  check_raw_levels(dat, "SMQ040", "^(1|2|3|7|9)$|EVERY DAY|SOME DAYS|NOT AT ALL|REFUSED|DON'T KNOW|DONT
  check_raw_levels(dat, "DIQ010", "^(1|2|3|7|9)$|YES|NO|BORDERLINE|REFUSED|DON'T KNOW|DONT KNOW"),
  check_raw_levels(dat, "BPQ020", "^(1|2|7|9)$|YES|NO|REFUSED|DON'T KNOW|DONT KNOW")
)
kable(harmonization_checks, caption = "Codebook-aligned raw coding checks by cycle. Non-zero unexpected
```

Table 12: Codebook-aligned raw coding checks by cycle. Non-zero unexpected counts indicate coding values not covered by harmonization rules.

variable	cycle	n_nonmissing	n_unexpected	unexpected_examples
RIAGENDR	2003-2004	10122	0	
RIAGENDR	2005-2006	10348	0	
RIAGENDR	2007-2008	10149	0	

variable	cycle	n_nonmissing	n_unexpected	unexpected_examples
RIAGENDR	2009-2010	10537	0	
RIAGENDR	2011-2012	9756	0	
RIAGENDR	2013-2014	10175	0	
RIAGENDR	2015-2016	9971	0	
RIAGENDR	2017-2018	9254	0	
DMDEDUC2	2003-2004	5041	0	
DMDEDUC2	2005-2006	4979	0	
DMDEDUC2	2007-2008	5935	0	
DMDEDUC2	2009-2010	6218	0	
DMDEDUC2	2011-2012	5560	0	
DMDEDUC2	2013-2014	5769	0	
DMDEDUC2	2015-2016	5719	0	
DMDEDUC2	2017-2018	5569	0	
SMQ040	2003-2004	2496	0	
SMQ040	2005-2006	2348	0	
SMQ040	2007-2008	2802	0	
SMQ040	2009-2010	2866	0	
SMQ040	2011-2012	2369	0	
SMQ040	2013-2014	2579	0	
SMQ040	2015-2016	2422	0	
SMQ040	2017-2018	2359	0	
DIQ010	2003-2004	9645	0	
DIQ010	2005-2006	9822	0	
DIQ010	2007-2008	9666	0	
DIQ010	2009-2010	10109	0	
DIQ010	2011-2012	9363	0	
DIQ010	2013-2014	9769	0	
DIQ010	2015-2016	9575	0	
DIQ010	2017-2018	8897	0	
BPQ020	2003-2004	6138	0	
BPQ020	2005-2006	6139	0	
BPQ020	2007-2008	6546	0	
BPQ020	2009-2010	6889	0	
BPQ020	2011-2012	6174	0	
BPQ020	2013-2014	6464	0	
BPQ020	2015-2016	6327	0	
BPQ020	2017-2018	6161	0	

```r
# --- CRP unit detection and conversion ---
# PhenoAge (Levine 2018) uses CRP in mg/dL. NHANES LBXCRP is typically mg/dL;
# LBXHSCRP or CRP tables in some cycles may report mg/L.
# Step 1: check variable/table provenance.
crp_by_cycle <- dat %>%
  group_by(cycle) %>%
  summarise(
    source_table = if ("crp_table" %in% names(pick(everything()))) paste(unique(na.omit(crp_table)), co
    n_obs = sum(!is.na(crp)),
    median_crp = median(crp, na.rm = TRUE),
    p75 = quantile(crp, 0.75, na.rm = TRUE),
    p90 = quantile(crp, 0.90, na.rm = TRUE),
    .groups = "drop"
  )
```

```r
kable(crp_by_cycle, caption = "CRP diagnostic by cycle -- verify units (PhenoAge expects mg/dL). If sou
```

Table 13: CRP diagnostic by cycle – verify units (PhenoAge expects mg/dL). If source table is HSCRP, values may be mg/L.

cycle	source_table	n_obs	median_crp	p75	p90
2003-2004		0	NA	NA	NA
2005-2006	CRP_D	8172	0.120	0.38	0.890
2007-2008	CRP_E	6809	0.160	0.40	0.920
2009-2010	CRP_F	8299	0.130	0.35	0.830
2011-2012		0	NA	NA	NA
2013-2014		0	NA	NA	NA
2015-2016	HSCRP_I	7867	1.200	3.60	8.200
2017-2018	HSCRP_J	7250	1.355	3.59	7.462

```r
# Step 2: provenance-based detection (strongest signal).
crp_tables_used <- if ("crp_table" %in% names(dat)) unique(na.omit(dat$crp_table)) else character(0)
provenance_suggests_mgl <- length(crp_tables_used) > 0 && any(str_detect(crp_tables_used, regex("HSCRP"
# Step 3: distribution-based backup -- require median > 1.5 AND p75 > 3 to trigger conversion.
crp_overall_med <- median(dat$crp, na.rm = TRUE)
crp_overall_p75 <- quantile(dat$crp, 0.75, na.rm = TRUE)
distribution_suggests_mgl <- !is.na(crp_overall_med) && crp_overall_med > 1.5 &&
                             !is.na(crp_overall_p75) && crp_overall_p75 > 3

if (provenance_suggests_mgl || distribution_suggests_mgl) {
  reason <- if (provenance_suggests_mgl) {
    paste0("CRP source table(s) [", paste(crp_tables_used, collapse = ", "), "] suggest mg/L.")
  } else {
    paste0("Distribution heuristic: median=", round(crp_overall_med, 3), ", p75=", round(crp_overall_p7
  }
  message(">>> Converting CRP from mg/L to mg/dL (/ 10). Reason: ", reason)
  dat$crp <- dat$crp / 10
} else {
  message(">>> CRP assumed to be in mg/dL (median=", round(crp_overall_med, 4),
          ", p75=", round(crp_overall_p75, 4), "). Verify against NHANES documentation.")
}

# --- Glucose provenance diagnostic ---
# Coerce glu to numeric (nhanes_raw has character columns from bind_rows); needed for summarise and lat
dat$glu <- suppressWarnings(as.numeric(as.character(dat$glu)))
glu_by_cycle <- dat %>%
  group_by(cycle) %>%
  summarise(
    source_table = if ("glu_table" %in% names(pick(everything()))) paste(unique(na.omit(glu_table)), co
    n_glu = sum(!is.na(glu)),
    median_glu = median(glu, na.rm = TRUE),
    min_glu = min(glu, na.rm = TRUE),
    max_glu = max(glu, na.rm = TRUE),
    .groups = "drop"
  )
kable(glu_by_cycle, caption = "Glucose provenance by cycle. If source_table is a fasting component (GLU
```

Table 14: Glucose provenance by cycle. If source_table is a fasting component (GLU_*), use a compatible fasting subsample weight.

cycle	source_table	n_glu	median_glu	min_glu	max_glu
2003-2004	L10_C	6492	89	46	512
2005-2006	GLU_D	6434	90	42	513
2007-2008	GLU_E	6377	91	33	559
2009-2010	GLU_F	6860	91	34	777
2011-2012	GLU_G	5976	91	31	617
2013-2014	GLU_H	6553	93	49	577
2015-2016	GLU_I	6257	92	19	610
2017-2018	GLU_J	5901	92	47	626

```
glu_med <- median(dat$glu, na.rm = TRUE)
if (!is.na(glu_med) && glu_med < 6) {
  warning("Glucose median is < 6 -- this looks like mmol/L. Converting to mg/dL ( x 18.018).")
  dat$glu <- dat$glu * 18.018
}

# Warn based on stored table provenance, not just weight variable existence.
glu_tables_used <- if ("glu_table" %in% names(dat)) unique(na.omit(dat$glu_table)) else character(0)
glu_is_fasting <- length(glu_tables_used) > 0 && any(str_detect(glu_tables_used, regex("^GLU_", ignore_
if (glu_is_fasting) {
  message(">>> IMPORTANT: Glucose sourced from fasting component table(s) [",
          paste(glu_tables_used, collapse = ", "),
          "]. A compatible fasting subsample weight (e.g., WTSAF2YR) is required. ",
          "Using a non-fasting or otherwise incompatible weight can bias estimates. ",
          "Review the weight audit; override config$weight_type only if auto-selection does not resolve
} else if (length(glu_tables_used) > 0) {
  message(">>> Glucose source table(s): ", paste(glu_tables_used, collapse = ", "),
          ". Verify whether this is a fasting component.")
}

# ------------------------------------------------------------
# CLEAN SPECIAL NHANES MISSING CODES FOR LAB VARIABLES
# ------------------------------------------------------------
clean_nhanes_numeric <- function(x) {
  x <- suppressWarnings(as.numeric(as.character(x)))
  special_codes <- c(
    7, 77, 777, 7777, 77777,
    9, 99, 999, 9999, 99999,
    888, 8888, 88888
  )
  x[x %in% special_codes] <- NA_real_
  x
}

dat$alb <- clean_nhanes_numeric(dat$alb)
dat$scr <- clean_nhanes_numeric(dat$scr)
dat$glu <- clean_nhanes_numeric(dat$glu)
dat$crp <- clean_nhanes_numeric(dat$crp)
dat$lym <- clean_nhanes_numeric(dat$lym)
dat$mcv <- clean_nhanes_numeric(dat$mcv)
```

```r
dat$rdw <- clean_nhanes_numeric(dat$rdw)
dat$alp <- clean_nhanes_numeric(dat$alp)
dat$wbc <- clean_nhanes_numeric(dat$wbc)

# ------------------------------------------------------------
# HARD PLAUSIBILITY CHECK (retain CRP >1 mg/dL in primary; flag as acute)
# ------------------------------------------------------------
dat <- dat %>%
  mutate(
    crp_acute = ifelse(!is.na(crp) & crp > 1, 1L, 0L),
    wbc = ifelse(wbc > 50, NA_real_, wbc),
    glu = ifelse(glu > 600, NA_real_, glu),
    alp = ifelse(alp > 500, NA_real_, alp)
  )

# --- General biomarker plausibility ---
biomarker_ranges <- list(
  alb  = c(1, 7,    "g/dL"),
  scr  = c(0.1, 25, "mg/dL"),
  glu  = c(20, 600,  "mg/dL"),
  crp  = c(0.001, 3, "mg/dL (post-conversion if applied)"),
  lym  = c(1, 80,    "%"),
  mcv  = c(50, 150,  "fL"),
  rdw  = c(8, 30,    "%"),
  alp  = c(5, 500,   "U/L"),
  wbc  = c(1, 50,    "1000 cells/uL")
)
biomarker_check <- lapply(names(biomarker_ranges), function(nm) {
  vals <- dat[[nm]]
  rng  <- biomarker_ranges[[nm]]
  lo   <- as.numeric(rng[1]); hi <- as.numeric(rng[2]); unit <- rng[3]
  v    <- vals[!is.na(vals)]
  n_out <- sum(v < lo | v > hi)
  tibble(biomarker = nm, unit = unit, n_obs = length(v),
         min = if (length(v) > 0) min(v) else NA_real_,
         p25 = if (length(v) > 0) quantile(v, 0.25) else NA_real_,
         median = if (length(v) > 0) median(v) else NA_real_,
         p75 = if (length(v) > 0) quantile(v, 0.75) else NA_real_,
         max = if (length(v) > 0) max(v) else NA_real_,
         expected_lo = lo, expected_hi = hi,
         n_out_of_range = n_out)
}) %>% bind_rows()
kable(biomarker_check, caption = "PhenoAge biomarker ranges after unit conversion (verify units)")
```

Table 15: PhenoAge biomarker ranges after unit conversion (verify units)

biomarker	unit	n_obs	min	p25	median	p75	max	expected_lo	expected_hi	n_out_of_range
alb	g/dL	50853	1.200	4.000	4.300	4.500	5.60	1e+00	7	0
scr	mg/dL	50848	0.140	0.700	0.820	0.990	17.80	1e-01	25	0
glu	mg/dL	49165	19.000	84.000	91.000	102.000	577.00	2e+01	600	1

biomarker	unit	n_obs	min	p25	median	p75	max	expected_lo	expected_hi	out_of_range
crp	mg/dL (post-conversion if applied)	38397	0.001	0.008	0.032	0.116	18.85	1e-03	3	197
lym	%	65864	2.600	26.300	32.500	39.900	94.50	1e+00	80	21
mcv	fL	65903	35.400	83.600	87.700	91.400	125.30	5e+01	150	1
rdw	%	65992	6.300	12.400	12.900	13.600	37.80	8e+00	30	7
alp	U/L	49756	10.000	56.000	70.000	90.000	500.00	5e+00	500	0
wbc	1000 cells/uL	64614	1.400	5.700	6.900	8.400	44.80	1e+00	50	0

```r
for (nm in names(biomarker_ranges)) {
  v <- dat[[nm]][!is.na(dat[[nm]])]
  lo <- as.numeric(biomarker_ranges[[nm]][1])
  hi <- as.numeric(biomarker_ranges[[nm]][2])
  pct_out <- if (length(v) > 0) mean(v < lo | v > hi) else 0
  if (pct_out > 0.20) warning(sprintf(
    "Biomarker '%s': %.1f%% of values outside plausible range [%s, %s]. Check units.",
    nm, pct_out * 100, lo, hi
  ))
}

# Recompute PhenoAge after any unit conversions (CRP mg/L->mg/dL, glucose mmol/L->mg/dL).
pheno_core <- compute_phenoage_components(dat, include_scr = TRUE, include_alb = TRUE)
dat$phenoage <- pheno_core$phenoage

message(
  "PhenoAge diagnostics -- xb median=", round(stats::median(pheno_core$xb, na.rm = TRUE), 3),
  ", mort median=", format(stats::median(pheno_core$mort, na.rm = TRUE), scientific = TRUE, digits = 3)
  ", pct(mort>0.9999)=", round(mean(pheno_core$mort > 0.9999, na.rm = TRUE) * 100, 1), "%",
  ", PhenoAge SD=", round(stats::sd(dat$phenoage, na.rm = TRUE), 3)
)
ph_q <- stats::quantile(dat$phenoage, probs = c(0, 0.25, 0.5, 0.75, 1), na.rm = TRUE)
message(
  "PhenoAge five-number summary -- min=", round(ph_q[[1]], 3),
  ", p25=", round(ph_q[[2]], 3),
  ", median=", round(ph_q[[3]], 3),
  ", p75=", round(ph_q[[4]], 3),
  ", max=", round(ph_q[[5]], 3)
)
if (mean(pheno_core$mort > 0.9999, na.rm = TRUE) > 0.25) {
  warning("More than 25% of participants have mort > 0.9999 after stabilization. Re-check biomarker uni
}
```

Exposure Processing and LOD Handling

```r
pfas_vars <- c("LBXPFOS", "LBXPFOA", "LBXPFNA", "LBXPFHS", "LBXPFDA")

# --- ISOMER HARMONIZATION (2013-2018) ---
# NHANES split PFOS and PFOA into linear and branched isomers.
# We must apply the LOD/sqrt(2) rule to the individual isomers FIRST, then sum them.
```

```r
isomer_map <- list(
  pfos_lin = c("SSNPFOS", "LBXNFOS"),
  pfos_br  = c("SSMPFOS", "LBXMFOS"),
  pfoa_lin = c("SSNPFOA", "LBXNFOA"),
  pfoa_br  = c("SSBPFOA", "LBXBFOA")
)

lc_map <- list(
  pfos_lin = c("SDNPFOSL", "LBDNFOSL"),
  pfos_br  = c("SDMPFOSL", "LBDMFOSL"),
  pfoa_lin = c("SDNPFOAL", "LBDNFOAL"),
  pfoa_br  = c("SDBPFOAL", "LBDBFOAL")
)

for (chem in c("pfos", "pfoa")) {
  # Get available variables for the cycle
  lin_vars <- isomer_map[[paste0(chem, "_lin")]]
  br_vars  <- isomer_map[[paste0(chem, "_br")]]
  lin_lc   <- lc_map[[paste0(chem, "_lin")]]
  br_lc    <- lc_map[[paste0(chem, "_br")]]

  lin_val <- coalesce_num(dat, lin_vars)
  br_val  <- coalesce_num(dat, br_vars)
  lin_lc_val <- coalesce_num(dat, lin_lc)
  br_lc_val  <- coalesce_num(dat, br_lc)

  # Apply LOD rule: if indicator == 1, divide by sqrt(2)
  lin_val <- ifelse(!is.na(lin_lc_val) & lin_lc_val == 1, lin_val / sqrt(2), lin_val)
  br_val  <- ifelse(!is.na(br_lc_val) & br_lc_val == 1, br_val / sqrt(2), br_val)

  # Sum them to recreate the total concentration
  summed_val <- lin_val + br_val

  # Inject back into the legacy total variables (LBXPFOS or LBXPFOA)
  target_var <- ifelse(chem == "pfos", "LBXPFOS", "LBXPFOA")

  if (target_var %in% names(dat)) {
    dat[[target_var]] <- coalesce_num(dat, c(target_var))
    dat[[target_var]] <- ifelse(is.na(dat[[target_var]]) & !is.na(summed_val), summed_val, dat[[target_
  } else {
    dat[[target_var]] <- summed_val
  }
}
# --- END ISOMER HARMONIZATION ---

blood_metal_vars <- c("LBXBPB", "LBXBCD", "LBXTHG")
urine_metal_vars <- c("URXUAS", "URXUCD", "URXUPB")

chosen_exposures <- switch(
  config$exposure_family,
  pfas = pfas_vars,
  metals = c(blood_metal_vars, urine_metal_vars),
  both = c(pfas_vars, blood_metal_vars, urine_metal_vars),
```

```r
  pfas_vars
)
chosen_exposures <- chosen_exposures[chosen_exposures %in% names(dat)]
if (length(chosen_exposures) < 3) stop("Too few exposure variables found; choose different exposure fam

for (v in chosen_exposures) {
  vv <- apply_lod_rule(dat, v)
  dat[[paste0(v, "_log2")]] <- log2(vv)
}

if (any(urine_metal_vars %in% chosen_exposures)) {
  for (uv in intersect(urine_metal_vars, chosen_exposures)) {
    vv <- apply_lod_rule(dat, uv)
    dat[[paste0(uv, "_ucr_adj_log2")]] <- log2((vv / pmax(dat$ucr, 1e-6)) * 100)
  }
}

exposure_vars <- names(dat)[str_detect(names(dat), "_log2$|_ucr_adj_log2$")]
exposure_vars <- unique(exposure_vars)
# Prevent collinear double-counting: use creatinine-adjusted urine metals only, drop raw _log2 for them
raw_urine_vars <- paste0(urine_metal_vars, "_log2")
exposure_vars <- setdiff(exposure_vars, raw_urine_vars)
# Primary analysis: fixed 4-PFAS subset common across all retained cycles. Sensitivity: available-expos
primary_exposure_vars <- c("LBXPFOS_log2", "LBXPFOA_log2", "LBXPFNA_log2", "LBXPFHS_log2")
primary_exposure_vars <- primary_exposure_vars[primary_exposure_vars %in% exposure_vars]
adaptive_exposure_vars <- exposure_vars

# LOD diagnostic table -- computed on dat (post-merge, pre-log transform) stratified by cycle.
lod_diag <- lapply(chosen_exposures, function(v) {
  x_raw <- suppressWarnings(as.numeric(as.character(dat[[v]])))
  base <- sub("^(LBX|URX)", "", v)
  lc_candidates <- c(
    paste0(v, "LC"), paste0("LBD", base, "LC"), paste0("URD", base, "LC"),
    paste0(v, "L"),  paste0("LBD", base, "L"),  paste0("URD", base, "L")
  )
  lc_var <- lc_candidates[lc_candidates %in% names(dat)]
  has_lc <- length(lc_var) > 0
  if (has_lc) {
    lc_vals <- suppressWarnings(as.numeric(as.character(dat[[lc_var[1]]])))
    below <- !is.na(lc_vals) & lc_vals == 1
  } else {
    below <- rep(FALSE, length(x_raw))
  }

  by_cycle <- dat %>%
    mutate(.x = x_raw, .below = below) %>%
    group_by(cycle) %>%
    summarise(
      n = sum(!is.na(.x)),
      n_below = sum(.below, na.rm = TRUE),
      pct_below = if (sum(!is.na(.x)) > 0) round(sum(.below, na.rm = TRUE) / sum(!is.na(.x)) * 100, 1) 
      median_below = if (sum(.below & !is.na(.x)) > 0) median(.x[.below], na.rm = TRUE) else NA_real_,
      median_above = if (sum(!.below & !is.na(.x)) > 0) median(.x[!.below & !is.na(.x)], na.rm = TRUE) 
```

```
      min_positive = if (any(.x > 0, na.rm = TRUE)) min(.x[.x > 0], na.rm = TRUE) else NA_real_,
      .groups = "drop"
    ) %>%
    mutate(exposure = v, lc_indicator = if (has_lc) lc_var[1] else "none") %>%
    select(exposure, cycle, lc_indicator, everything())
  by_cycle
}) %>% bind_rows()
kable(lod_diag, caption = "LOD diagnostic per exposure by cycle. Verify: median_below ~ LOD value; if i
```

Table 16: LOD diagnostic per exposure by cycle. Verify: median_below ~ LOD value; if it diverges from min_positive, review the LC->x/sqrt(2) assumption.

exposure	cycle	lc_indicator	n	n_below	pct_below	median_below	median_above	min_positive
LBXPFOS	2003-2004	LBDPFOSL	2094	0	0	NA	19.900	0.300
LBXPFOS	2005-2006	LBDPFOSL	2120	0	0	NA	15.900	0.140
LBXPFOS	2007-2008	LBDPFOSL	2100	0	0	NA	13.400	0.140
LBXPFOS	2009-2010	LBDPFOSL	2233	0	0	NA	9.000	0.140
LBXPFOS	2011-2012	LBDPFOSL	1904	0	0	NA	6.425	0.140
LBXPFOS	2013-2014	LBDPFOSL	0	0	NA	NA	NA	NA
LBXPFOS	2015-2016	LBDPFOSL	1993	0	0	NA	4.700	0.140
LBXPFOS	2017-2018	LBDPFOSL	1929	0	0	NA	4.200	0.140
LBXPFOA	2003-2004	LBDPFOAL	2094	0	0	NA	3.800	0.100
LBXPFOA	2005-2006	LBDPFOAL	2120	0	0	NA	3.700	0.070
LBXPFOA	2007-2008	LBDPFOAL	2100	0	0	NA	4.200	0.070
LBXPFOA	2009-2010	LBDPFOAL	2233	0	0	NA	2.900	0.070
LBXPFOA	2011-2012	LBDPFOAL	1904	0	0	NA	2.050	0.070
LBXPFOA	2013-2014	LBDPFOAL	0	0	NA	NA	NA	NA
LBXPFOA	2015-2016	LBDPFOAL	1993	0	0	NA	1.470	0.140
LBXPFOA	2017-2018	LBDPFOAL	1929	0	0	NA	1.370	0.140
LBXPFNA	2003-2004	LBDPFNAL	2094	0	0	NA	0.900	0.100
LBXPFNA	2005-2006	LBDPFNAL	2120	0	0	NA	1.000	0.070
LBXPFNA	2007-2008	LBDPFNAL	2100	0	0	NA	1.148	0.058

exposure	cycle	lc_indicator	n	n_below	pct_below	median_below	median_above	min_positive
LBXPFNA	2009-2010	LBDPFNAL	2233	0	0	NA	1.148	0.058
LBXPFNA	2011-2012	LBDPFNAL	1904	0	0	NA	0.890	0.060
LBXPFNA	2013-2014	LBDPFNAL	2168	0	0	NA	0.700	0.070
LBXPFNA	2015-2016	LBDPFNAL	1993	0	0	NA	0.600	0.070
LBXPFNA	2017-2018	LBDPFNAL	1929	0	0	NA	0.400	0.070
LBXPFHS	2003-2004	LBDPFHSL	2094	0	0	NA	1.900	0.200
LBXPFHS	2005-2006	LBDPFHSL	2120	0	0	NA	1.700	0.070
LBXPFHS	2007-2008	LBDPFHSL	2100	0	0	NA	2.000	0.070
LBXPFHS	2009-2010	LBDPFHSL	2233	0	0	NA	1.600	0.070
LBXPFHS	2011-2012	LBDPFHSL	1904	0	0	NA	1.190	0.070
LBXPFHS	2013-2014	LBDPFHSL	2168	0	0	NA	1.300	0.070
LBXPFHS	2015-2016	LBDPFHSL	1993	0	0	NA	1.200	0.070
LBXPFHS	2017-2018	LBDPFHSL	1929	0	0	NA	1.100	0.070

Analytic Sample and Survey Design

```
analytic <- dat %>%
  filter(age >= config$min_age)

# PFAS workflow: harmonize rotating 2-year subsample weights into one unified PFAS weight (WTPFAS2YR).
analytic <- analytic %>%
  mutate(
    WTPFAS2YR = coalesce_num(cur_data(), c("WTSA2YR", "WTSB2YR", "WTSC2YR", "WTSAF2YR"))
  )

weight_candidates <- names(analytic)[stringr::str_detect(names(analytic), "^WT")]
kable(data.frame(weight_variable = weight_candidates), caption = "Weight audit: all WT* variables in ana
```

Table 17: Weight audit: all WT* variables in analytic dataset.

weight_variable
WTINT2YR
WTMEC2YR
WTSA2YR
WTSAF2YR

weight_variable
WTSB2YR
WTSC2YR
WTPFAS2YR

```
# Exposure-specific weight audit in the exposure-complete analytic sample (fixed primary PFAS set).
complete_expo <- analytic %>%
  mutate(.has_all_expo = if_all(all_of(primary_exposure_vars), ~ !is.na(.x)))
wt_cols <- names(analytic)[str_detect(names(analytic), "^WT")]
# PFAS auto-selection eligibility: exclude full-sample interview/MEC weights; keep PFAS-appropriate sub
if (tolower(config$exposure_family) == "pfas") {
  wt_cols <- setdiff(wt_cols, c("WTINT2YR", "WTMEC2YR"))
}
if (length(wt_cols) > 0) {
  wt_audit <- complete_expo %>%
    filter(.has_all_expo) %>%
    summarise(across(all_of(wt_cols), ~ round(mean(!is.na(.x)) * 100, 1))) %>%
    pivot_longer(everything(), names_to = "weight", values_to = "pct_nonmissing_complete_expo") %>%
    arrange(desc(pct_nonmissing_complete_expo))
  kable(wt_audit, caption = "Weight availability (%) among participants with complete exposure set. Use
}
```

Table 18: Weight availability (%) among participants with complete exposure set. Use the weight with highest coverage that matches your subsample.

weight	pct_nonmissing_complete_expo
WTPFAS2YR	100.0
WTSAF2YR	42.6
WTSA2YR	39.6
WTSC2YR	31.8
WTSB2YR	28.7

```
# Auto-weight: rank eligible candidate weights by coverage and apply the top PFAS-appropriate subsample
if (config$weight_type == "auto" && exists("wt_audit") && nrow(wt_audit) > 0) {
  wt_audit_auto <- wt_audit
  if (tolower(config$exposure_family) == "pfas") {
    wt_audit_auto <- wt_audit_auto %>% filter(!(weight %in% c("WTINT2YR", "WTMEC2YR")))
  }
  if (nrow(wt_audit_auto) == 0) {
    stop("Auto-weight selection failed for PFAS: no eligible WTS* subsample weights were available in t
  }
  top_row <- wt_audit_auto %>% slice(1)
  config$weight_type <- top_row$weight
  message(">>> Auto-weight resolved: Applying '", config$weight_type, "' (Coverage: ", top_row$pct_nonm
}

weight_info <- resolve_weight_var(analytic, config$weight_type, config$exposure_family)

analytic$weight_2yr <- suppressWarnings(as.numeric(as.character(analytic[[weight_info$var]])))
```

```r
analytic <- analytic %>%
  filter(!is.na(phenoage), !is.na(age), !is.na(ckd), !is.na(weight_2yr), !is.na(strata), !is.na(psu))

n_cycles <- length(unique(analytic$cycle))
# After resolving the PFAS-appropriate 2-year weight, pooled analyses divide by the number of contribut
n_cycles_exposure_available <- analytic %>%
  filter(if_any(all_of(primary_exposure_vars), ~ !is.na(.x))) %>%
  summarise(n = n_distinct(cycle)) %>%
  pull(n)
if (!is.finite(n_cycles_exposure_available) || n_cycles_exposure_available < 1) {
  n_cycles_exposure_available <- n_cycles
}
repool_weights <- function(df) {
  ncy <- dplyr::n_distinct(df$cycle[!is.na(df$weight_2yr)])
  if (!is.finite(ncy) || ncy < 1) stop("Invalid cycle count for weight pooling.")
  df$weight_combined <- df$weight_2yr / ncy
  df
}
analytic <- repool_weights(analytic)
message("Weight pooling: repool_weights() applied to analytic (n_cycles in sample = ", n_cycles_exposure

# Audit B: weight availability in the final analytic sample (after all exclusions).
if (length(wt_cols) > 0) {
  wt_audit_final <- analytic %>%
    summarise(across(any_of(wt_cols), ~ round(mean(!is.na(.x)) * 100, 1))) %>%
    pivot_longer(everything(), names_to = "weight", values_to = "pct_nonmissing_final_sample") %>%
    arrange(desc(pct_nonmissing_final_sample))
  kable(wt_audit_final, caption = "Weight availability (%) in final analytic sample (post-exclusion). T
}
```

Table 19: Weight availability (%) in final analytic sample (post-exclusion). This is the definitive audit.

weight	pct_nonmissing_final_sample
WTPFAS2YR	100.0
WTSAF2YR	74.1
WTSC2YR	22.1
WTSB2YR	19.5
WTSA2YR	9.0

```r
# PRIMARY: survey-weighted residualization of PhenoAge on age.
# This ensures PhenoAgeAccel is a population-representative construct.
analytic$phenoage_accel_unweighted <- NA_real_
fit_age_uw <- lm(phenoage ~ age, data = analytic)
analytic$phenoage_accel_unweighted <- residuals(fit_age_uw)
keep_w <- which(!is.na(analytic$phenoage) & !is.na(analytic$age) & !is.na(analytic$weight_combined))
if (length(keep_w) > 0) {
  svy_resid <- svydesign(id = ~psu, strata = ~strata, weights = ~weight_combined, nest = TRUE,
                         data = analytic[keep_w, , drop = FALSE])
  fit_age_wt <- svyglm(phenoage ~ age, design = svy_resid, family = gaussian())
  pred_wt <- as.numeric(predict(fit_age_wt))
  analytic$phenoage_accel <- NA_real_
```

```r
  analytic$phenoage_accel[keep_w] <- analytic$phenoage[keep_w] - pred_wt
}
analytic <- analytic %>% filter(!is.na(phenoage_accel))
message(">>> PhenoAgeAccel: primary = survey-weighted residualization; unweighted version retained as s

ckd_marker_availability <- tibble::tibble(
  group = c("Both eGFR and ACR", "eGFR only", "ACR only", "Neither"),
  N = c(
    sum(!is.na(analytic$egfr) & !is.na(analytic$acr)),
    sum(!is.na(analytic$egfr) & is.na(analytic$acr)),
    sum(is.na(analytic$egfr) & !is.na(analytic$acr)),
    sum(is.na(analytic$egfr) & is.na(analytic$acr))
  )
)

if (!all(c(0, 1) %in% unique(analytic$ckd))) {
  stop("Both CKD groups are not represented after exclusions.")
}

svy <- svydesign(
  id = ~psu, strata = ~strata, weights = ~weight_combined, nest = TRUE, data = analytic
)

kable(data.frame(weight_selected = weight_info$var, note = weight_info$note), caption = "Selected NHANES
```

Table 20: Selected NHANES weight variable

weight_selected	note
WTPFAS2YR	User-specified weight: WTPFAS2YR

```r
kable(ckd_marker_availability, caption = "CKD marker availability in analytic sample")
```

Table 21: CKD marker availability in analytic sample

group	N
Both eGFR and ACR	15019
eGFR only	220
ACR only	0
Neither	0

Mixture Modeling

```r
valid_num <- function(v, d) v %in% names(d) && length(unique(stats::na.omit(d[[v]]))) >= 2
valid_cat <- function(v, d) v %in% names(d) && length(unique(stats::na.omit(as.character(d[[v]])))) >= 

# Primary adjustment: lean set to avoid overadjustment/mediator bias (diabetes, HTN, CVD may be on path
# Aligned with EHP/Env Int mixture papers: sex, race, PIR, BMI, smoking, cycle only.
```

```r
clinical_covars <- c("diabetes", "hypertension", "cvd")  # used only in sensitivity "cardiometabolic"
# No age: outcome is phenoage_accel (age-residualized). Same 6 covariates for qgcomp and Model B.
covars <- c(
  if (valid_cat("sex", analytic)) "factor(sex)",
  if (valid_cat("race_eth", analytic)) "factor(race_eth)",
  if (valid_num("pir", analytic)) "pir",
  if (valid_num("smoker", analytic)) "smoker",
  if (valid_num("bmi", analytic)) "bmi",
  if (valid_cat("cycle", analytic)) "factor(cycle)"
)

# Sanity checks before modeling.
message("--- Pre-modeling sanity checks ---")
message("Analytic N: ", nrow(analytic))
message("PhenoAgeAccel: mean=", round(mean(analytic$phenoage_accel, na.rm = TRUE), 3),
        ", SD=", round(sd(analytic$phenoage_accel, na.rm = TRUE), 3))
message("CKD prevalence: ", round(mean(analytic$ckd == 1, na.rm = TRUE) * 100, 1), "%")
message("Weight selected: ", weight_info$var)

# qgcomp complete-case: use primary (fixed) exposure vars with sufficient variability and non-missing i
qg_min_frac <- 0.5
qg_exposure_vars_for_cc <- primary_exposure_vars[vapply(primary_exposure_vars, function(v) {
  x <- suppressWarnings(as.numeric(analytic[[v]]))
  finite <- is.finite(x)
  n_ok <- sum(finite)
  pct_ok <- n_ok / max(1, nrow(analytic))
  n_ok > 0 && dplyr::n_distinct(x[finite]) >= 2 && pct_ok >= qg_min_frac
}, logical(1))]
if (length(qg_exposure_vars_for_cc) < 2) {
  qg_exposure_vars_for_cc <- primary_exposure_vars[vapply(primary_exposure_vars, function(v) {
    x <- suppressWarnings(as.numeric(analytic[[v]]))
    x <- x[is.finite(x)]
    length(x) > 0 && dplyr::n_distinct(x) >= 2
  }, logical(1))]
  if (length(qg_exposure_vars_for_cc) < 2) qg_exposure_vars_for_cc <- primary_exposure_vars
  message("qgcomp: few primary exposures met variance + ", round(qg_min_frac * 100), "% non-missing; usi
}
message("qgcomp exposures for complete-case (primary fixed set, variance + >= ", round(qg_min_frac * 10
# Qgcomp uses same 6-variable adjustment as Model B (sex, race, PIR, BMI, smoking, cycle); no age (outc
rhs_qg_cc <- paste(c(qg_exposure_vars_for_cc, covars), collapse = " + ")
f_qg_cc <- as.formula(paste("phenoage_accel ~", rhs_qg_cc))
qg_cc <- complete.cases(model.frame(f_qg_cc, data = analytic, na.action = na.pass))
analytic_qg <- analytic[qg_cc, , drop = FALSE]
analytic_qg <- repool_weights(analytic_qg)
# Normalize precision weights for qgcomp so glm does not treat sum(weights) as N (avoids N = 300M, infi
analytic_qg$weight_normalized <- analytic_qg$weight_combined / mean(analytic_qg$weight_combined, na.rm =
# Ensure covariates are proper type for qgcomp (character -> wrong model matrix, psi ~ 0).
for (v in c("phenoage_accel", "age", "pir", "smoker", "bmi")) {
  if (v %in% names(analytic_qg)) analytic_qg[[v]] <- suppressWarnings(as.numeric(as.character(analytic_
}
analytic_qg$sex      <- factor(as.character(analytic_qg$sex))
analytic_qg$race_eth <- factor(as.character(analytic_qg$race_eth))
analytic_qg$cycle    <- factor(as.character(analytic_qg$cycle))
```

```r
for (v in qg_exposure_vars_for_cc) {
  if (v %in% names(analytic_qg)) analytic_qg[[v]] <- suppressWarnings(as.numeric(as.character(analytic_
}
message("qgcomp complete-case: same 6 covars as Model B; N = ", nrow(analytic_qg))

# Among those, keep only exposures that still have variance in the complete-case sample (in case one co
qg_exposure_vars <- qg_exposure_vars_for_cc[vapply(qg_exposure_vars_for_cc, function(v) {
  x <- suppressWarnings(as.numeric(analytic_qg[[v]]))
  x <- x[is.finite(x)]
  length(x) > 0 && dplyr::n_distinct(x) >= 2
}, logical(1))]
if (length(qg_exposure_vars) < 2) {
  warning("Too few finite/variable exposures for qgcomp after complete-case filtering; Table 2 will be
}
if (length(qg_exposure_vars) > 0) {
  analytic_qg <- analytic_qg %>%
    filter(if_all(all_of(c("phenoage_accel", qg_exposure_vars)), ~ is.finite(.x)))
}
q_qg <- as.integer(config$q)
if (!is.finite(q_qg) || q_qg < 2) stop("config$q must be an integer >= 2.")

# Qgcomp model: same 6-variable adjustment as Model B (no age).
covars_qg <- c(
  if (valid_cat("sex", analytic_qg)) "factor(sex)",
  if (valid_cat("race_eth", analytic_qg)) "factor(race_eth)",
  if (valid_num("pir", analytic_qg)) "pir",
  if (valid_num("smoker", analytic_qg)) "smoker",
  if (valid_num("bmi", analytic_qg)) "bmi",
  if (valid_cat("cycle", analytic_qg)) "factor(cycle)"
)
qg_covars_intended <- covars_qg
build_qg_formula <- function(exps, covs) {
  rhs <- paste(c(exps, covs), collapse = " + ")
  as.formula(paste("phenoage_accel ~", rhs))
}
f_qg <- build_qg_formula(qg_exposure_vars, covars_qg)

repeat {
  mf_qg <- tryCatch(model.frame(f_qg, data = analytic_qg, na.action = na.omit), error = function(e) NULL
  if (is.null(mf_qg) || nrow(mf_qg) == 0) break
  factor_terms_qg <- names(mf_qg)[vapply(mf_qg, is.factor, logical(1))]
  bad_factor_terms_qg <- factor_terms_qg[vapply(mf_qg[factor_terms_qg], function(x) nlevels(droplevels(
  removable_qg <- intersect(covars_qg, bad_factor_terms_qg)
  if (length(removable_qg) == 0) break
  message("qgcomp overall: dropping collapsed factor term(s): ", paste(removable_qg, collapse = ", "))
  covars_qg <- setdiff(covars_qg, removable_qg)
  f_qg <- build_qg_formula(qg_exposure_vars, covars_qg)
}
qg_covars_realized <- covars_qg
message("qgcomp exposures used: ", paste(qg_exposure_vars, collapse = ", "))
message("qgcomp finite analytic N: ", nrow(analytic_qg))

qgcomp_overall_error <- NULL
```

```r
fit_qg_overall <- tryCatch(
  qgcomp::qgcomp.noboot(
    f = f_qg, expnms = qg_exposure_vars, data = analytic_qg,
    family = gaussian(), q = q_qg, weights = analytic_qg$weight_normalized
  ),
  error = function(e) {
    qgcomp_overall_error <<- conditionMessage(e)
    NULL
  }
)
if (!is.null(fit_qg_overall)) {
  message("qgcomp N used: ", fit_qg_overall$fit$n[1])
} else {
  warning("qgcomp overall model failed: ", qgcomp_overall_error)
}

# Survey-weighted interaction model using quantized summed score.
# Quantile cutpoints are computed within each NHANES cycle using cycle-specific sampling weights.
qdat <- analytic
qdat$ckd <- as.integer(qdat$ckd)
# Strict sample = complete-case on qgcomp-compatible PFAS subset (same as analytic_qg). Label reflects
qdat_strict <- analytic_qg
qdat_strict$ckd <- as.integer(qdat_strict$ckd)

# Primary model: fixed PFAS set (primary_exposure_vars). Restrict to complete case on primary set.
qdat <- qdat %>% filter(if_all(all_of(primary_exposure_vars), ~ !is.na(.x)))
if (config$require_complete_exposure_set) {
  # Optional extra: could also require full adaptive set here; we use primary only for main.
}
message("Primary interaction: fixed PFAS set = ", paste(primary_exposure_vars, collapse = ", "))
message("N primary pre-quantile sample: ", nrow(qdat))
message("N strict sample (Strict complete-case, qgcomp-compatible PFAS subset): ", nrow(qdat_strict))

weighted_quantile <- function(x, w, probs) {
  keep <- is.finite(x) & is.finite(w) & w > 0
  x <- x[keep]; w <- w[keep]
  if (length(x) == 0) return(rep(NA_real_, length(probs)))
  ord <- order(x)
  x <- x[ord]; w <- w[ord]
  cw <- cumsum(w) / sum(w)
  sapply(probs, function(p) {
    idx <- which(cw >= p)[1]
    if (is.na(idx)) x[length(x)] else x[idx]
  })
}

# Primary mixture score: fixed set, cycle-specific quantiles to avoid temporal confounding (PFAS trends
for (v in primary_exposure_vars) {
  qdat[[paste0(v, "_q")]] <- NA_real_
  for (cy in unique(qdat$cycle)) {
    idx <- qdat$cycle == cy
    x_cy <- suppressWarnings(as.numeric(qdat[[v]][idx]))
    w_cy <- suppressWarnings(as.numeric(qdat$weight_combined[idx]))
```

```r
    probs <- weighted_quantile(x_cy, w_cy, probs = seq(0, 1, length.out = config$q + 1))
    br <- sort(unique(as.numeric(probs)))
    if (length(br) > 1) {
      br[1] <- -Inf
      br[length(br)] <- Inf
    }
    if (length(br) < 3 || any(diff(br) <= 0)) {
      r <- rank(qdat[[v]][idx], na.last = "keep", ties.method = "average")
      qdat[[paste0(v, "_q")]][idx] <- as.numeric(cut(r, breaks = config$q, labels = FALSE, include.lowe
    } else {
      qdat[[paste0(v, "_q")]][idx] <- as.numeric(cut(qdat[[v]][idx], breaks = br, include.lowest = TRUE
    }
  }
}
qvars <- paste0(primary_exposure_vars, "_q")
qdat$nonmissing_q_exposures <- rowSums(!is.na(qdat[, qvars, drop = FALSE]))
qdat$mixture_qscore <- rowMeans(qdat[, qvars, drop = FALSE], na.rm = TRUE)
# Primary: require all primary PFAS non-missing (fixed set).
required_primary <- length(primary_exposure_vars)
qdat$mixture_qscore <- ifelse(qdat$nonmissing_q_exposures < required_primary, NA_real_, qdat$mixture_qs
# By-cycle table for reporting (primary set is fixed; this documents it).
cycle_capacity <- tibble(
  cycle = unique(qdat$cycle),
  measured_exposures = length(primary_exposure_vars),
  PFAS_measured = paste(primary_exposure_vars, collapse = ", "),
  required_nonmissing_exposures = required_primary
)

# Build strict mixture score: same exposure set as qgcomp, cycle-specific quantiles.
for (v in qg_exposure_vars) {
  qdat_strict[[paste0(v, "_q")]] <- NA_real_
  for (cy in unique(qdat_strict$cycle)) {
    idx <- qdat_strict$cycle == cy
    x_cy <- suppressWarnings(as.numeric(qdat_strict[[v]][idx]))
    w_cy <- suppressWarnings(as.numeric(qdat_strict$weight_combined[idx]))
    probs_s <- weighted_quantile(x_cy, w_cy, probs = seq(0, 1, length.out = config$q + 1))
    br_s <- sort(unique(as.numeric(probs_s)))
    if (length(br_s) > 1) {
      br_s[1] <- -Inf
      br_s[length(br_s)] <- Inf
    }
    if (length(br_s) < 3 || any(diff(br_s) <= 0)) {
      r_s <- rank(qdat_strict[[v]][idx], na.last = "keep", ties.method = "average")
      qdat_strict[[paste0(v, "_q")]][idx] <- as.numeric(cut(r_s, breaks = config$q, labels = FALSE, incl
    } else {
      qdat_strict[[paste0(v, "_q")]][idx] <- as.numeric(cut(qdat_strict[[v]][idx], breaks = br_s, inclu
    }
  }
}
qvars_strict <- paste0(qg_exposure_vars, "_q")
qdat_strict$nonmissing_q_exposures <- rowSums(!is.na(qdat_strict[, qvars_strict, drop = FALSE]))
qdat_strict$mixture_qscore <- rowMeans(qdat_strict[, qvars_strict, drop = FALSE], na.rm = TRUE)
```

```r
# Hard type enforcement to prevent silent coercion issues in svyglm.
qdat <- qdat %>%
  mutate(
    phenoage_accel = suppressWarnings(as.numeric(phenoage_accel)),
    mixture_qscore = suppressWarnings(as.numeric(mixture_qscore)),
    weight_combined = suppressWarnings(as.numeric(weight_combined)),
    psu = suppressWarnings(as.numeric(psu)),
    strata = suppressWarnings(as.numeric(strata)),
    ckd = as.integer(ckd)
  )
qdat_strict <- qdat_strict %>%
  mutate(
    phenoage_accel = suppressWarnings(as.numeric(phenoage_accel)),
    mixture_qscore = suppressWarnings(as.numeric(mixture_qscore)),
    weight_combined = suppressWarnings(as.numeric(weight_combined)),
    psu = suppressWarnings(as.numeric(psu)),
    strata = suppressWarnings(as.numeric(strata)),
    ckd = as.integer(ckd)
  )

# Primary model: fixed PFAS set (same in every cycle).
cycle_capacity_with_min <- cycle_capacity
kable(
  cycle_capacity_with_min,
  caption = "Primary analysis: fixed PFAS set by cycle. Same chemicals and required non-missing in all c
)
```

Table 22: Primary analysis: fixed PFAS set by cycle. Same chemicals and required non-missing in all cycles.

cycle	measured_exposures	PFAS_measured	required_nonmissing_exposures
2005-2006	4	LBXPFOS_log2, LBXPFOA_log2, LBXPFNA_log2, LBXPFHS_log2	4
2007-2008	4	LBXPFOS_log2, LBXPFOA_log2, LBXPFNA_log2, LBXPFHS_log2	4
2009-2010	4	LBXPFOS_log2, LBXPFOA_log2, LBXPFNA_log2, LBXPFHS_log2	4
2015-2016	4	LBXPFOS_log2, LBXPFOA_log2, LBXPFNA_log2, LBXPFHS_log2	4
2017-2018	4	LBXPFOS_log2, LBXPFOA_log2, LBXPFNA_log2, LBXPFHS_log2	4

```r
kable(
  tibble(
    statistic = c("N", "Mean", "SD", "Min", "P25", "Median", "P75", "Max"),
    value = c(
      sum(!is.na(qdat$mixture_qscore)),
      round(mean(qdat$mixture_qscore, na.rm = TRUE), 3),
      round(sd(qdat$mixture_qscore, na.rm = TRUE), 3),
      round(min(qdat$mixture_qscore, na.rm = TRUE), 3),
      round(quantile(qdat$mixture_qscore, 0.25, na.rm = TRUE), 3),
      round(median(qdat$mixture_qscore, na.rm = TRUE), 3),
```

```
      round(quantile(qdat$mixture_qscore, 0.75, na.rm = TRUE), 3),
      round(max(qdat$mixture_qscore, na.rm = TRUE), 3)
    )
  ),
  caption = paste0("Primary mixture quantile score (fixed set, N chemicals = ", length(primary_exposure
                   "). Score = mean quantile bin across ", paste(primary_exposure_vars, collapse = ", "
                   ". Complete case on primary set required.")
)
```

Table 23: Primary mixture quantile score (fixed set, N chemicals = 4). Score = mean quantile bin across LBXPFOS_log2, LBXPFOA_log2, LBXPFNA_log2, LBXPFHS_log2. Complete case on primary set required.

statistic	value
N	7633.000
Mean	2.419
SD	0.933
Min	1.000
P25	1.500
Median	2.500
P75	3.250
Max	4.000

```
stopifnot(all(c("psu", "strata", "weight_combined") %in% names(qdat)))
qdat <- qdat %>% filter(!is.na(psu), !is.na(strata), !is.na(weight_combined), !is.na(phenoage_accel), !
qdat_strict <- qdat_strict %>% filter(!is.na(psu), !is.na(strata), !is.na(weight_combined), !is.na(phen
qdat         <- repool_weights(qdat)
qdat_strict  <- repool_weights(qdat_strict)

# Ensure all model variables are proper type so svyglm does not zero out coefficients (character -> wro
for (v in exposure_vars) {
  if (v %in% names(qdat)) qdat[[v]] <- suppressWarnings(as.numeric(as.character(qdat[[v]])))
}
qdat$phenoage_accel <- suppressWarnings(as.numeric(as.character(qdat$phenoage_accel)))
qdat$age            <- suppressWarnings(as.numeric(as.character(qdat$age)))
qdat$pir            <- suppressWarnings(as.numeric(as.character(qdat$pir)))
qdat$smoker         <- suppressWarnings(as.numeric(as.character(qdat$smoker)))
qdat$bmi            <- suppressWarnings(as.numeric(as.character(qdat$bmi)))
qdat$sex            <- factor(as.character(qdat$sex))
qdat$race_eth       <- factor(as.character(qdat$race_eth))
qdat$cycle          <- factor(as.character(qdat$cycle))
qdat$ckd            <- as.integer(qdat$ckd)

for (v in exposure_vars) {
  if (v %in% names(qdat_strict)) qdat_strict[[v]] <- suppressWarnings(as.numeric(as.character(qdat_stri
}
qdat_strict$phenoage_accel <- suppressWarnings(as.numeric(as.character(qdat_strict$phenoage_accel)))
qdat_strict$age            <- suppressWarnings(as.numeric(as.character(qdat_strict$age)))
qdat_strict$pir            <- suppressWarnings(as.numeric(as.character(qdat_strict$pir)))
qdat_strict$smoker         <- suppressWarnings(as.numeric(as.character(qdat_strict$smoker)))
qdat_strict$bmi            <- suppressWarnings(as.numeric(as.character(qdat_strict$bmi)))
```

```r
qdat_strict$sex            <- factor(as.character(qdat_strict$sex))
qdat_strict$race_eth       <- factor(as.character(qdat_strict$race_eth))
qdat_strict$cycle          <- factor(as.character(qdat_strict$cycle))
qdat_strict$ckd            <- as.integer(qdat_strict$ckd)

message("---- Model B diagnostics ----")
message("N qdat: ", nrow(qdat))
print(table(qdat$ckd, useNA = "ifany"))
```

```
##
##    0    1
## 6315 1318
```

```r
message("N qdat_strict: ", nrow(qdat_strict))
print(table(qdat_strict$ckd, useNA = "ifany"))
```

```
##
##    0    1
## 2473  585
```

```r
sd_all <- sd(qdat$mixture_qscore, na.rm = TRUE)
sd_by <- tapply(qdat$mixture_qscore, qdat$ckd, sd, na.rm = TRUE)
message("SD(mixture_qscore) overall: ", sd_all)
print(sd_by)
```

```
##         0         1
## 0.9268457 0.9599365
```

```r
if (length(unique(stats::na.omit(qdat$ckd))) < 2) {
  stop("After exposure/design filtering, only one CKD group remains in qdat. ",
       "Interaction mixture_qscore*ckd cannot be estimated. ",
       "Set config$require_complete_exposure_set = FALSE or reduce exposure set/cycles.")
}
if (any(is.finite(sd_by) & sd_by < 1e-8)) {
  stop("mixture_qscore has near-zero variance within at least one CKD group. ",
       "Interaction cannot be estimated; check quantile binning/ties and exposure variability.")
}

miss_compare_ckd <- qdat %>%
  mutate(
    retained_primary = complete.cases(phenoage_accel, mixture_qscore, ckd, age, sex, race_eth, pir, smo
  ) %>%
  group_by(ckd, retained_primary) %>%
  summarise(
    N = n(),
    mean_age = round(mean(age, na.rm = TRUE), 1),
    female_pct = round(mean(sex == "Female", na.rm = TRUE) * 100, 1),
    .groups = "drop"
  ) %>%
  group_by(ckd) %>%
  mutate(pct_within_ckd = round(100 * N / sum(N), 1)) %>%
```

```
  ungroup() %>%
  mutate(
    ckd = ifelse(ckd == 1, "CKD", "No CKD"),
    retained_primary = ifelse(retained_primary, "Retained in Model B", "Dropped from Model B")
  )
kable(miss_compare_ckd, caption = "Primary interaction model retention by CKD status (retained vs dropp
```

Table 24: Primary interaction model retention by CKD status (retained vs dropped participants).

ckd	retained_primary	N	mean_age	female_pct	pct_within_ckd
No CKD	Dropped from Model B	3842	46.0	58.8	60.8
No CKD	Retained in Model B	2473	48.7	41.2	39.2
CKD	Dropped from Model B	733	62.6	61.7	55.6
CKD	Retained in Model B	585	64.6	40.5	44.4

```
# Use explicit data-frame copy so survey design uses correct types (avoids ~0 coefficients from referen
qdat_svy <- as.data.frame(qdat)
qdat_svy$phenoage_accel <- as.numeric(qdat_svy$phenoage_accel)
qdat_svy$mixture_qscore <- as.numeric(qdat_svy$mixture_qscore)
qdat_svy$ckd            <- as.integer(qdat_svy$ckd)
qdat_svy$age            <- as.numeric(qdat_svy$age)
qdat_svy$pir            <- as.numeric(qdat_svy$pir)
qdat_svy$smoker         <- as.numeric(qdat_svy$smoker)
qdat_svy$bmi            <- as.numeric(qdat_svy$bmi)
qdat_svy$sex            <- factor(qdat_svy$sex)
qdat_svy$race_eth       <- factor(qdat_svy$race_eth)
qdat_svy$cycle          <- factor(qdat_svy$cycle)
qdat_svy$weight_combined <- as.numeric(qdat_svy$weight_combined)
qdat_svy$psu            <- as.numeric(qdat_svy$psu)
qdat_svy$strata         <- as.numeric(qdat_svy$strata)
# Drop unused factor levels to avoid singularity with full survey design.
qdat_svy$sex      <- droplevels(factor(qdat_svy$sex))
qdat_svy$race_eth <- droplevels(factor(qdat_svy$race_eth))
qdat_svy$cycle    <- droplevels(factor(qdat_svy$cycle))

# Full complex survey design (PSU/strata) for valid inference; droplevels above avoids empty cells.
svy_q <- tryCatch(
  svydesign(id = ~psu, strata = ~strata, weights = ~weight_combined, nest = TRUE, data = qdat_svy),
  error = function(e) {
    message("Full design failed (", conditionMessage(e), "); falling back to weights-only design.")
    svydesign(id = ~1, strata = NULL, weights = ~weight_combined, data = qdat_svy)
  }
)
svy_q_strict <- if (nrow(qdat_strict) > 0) {
  qdat_strict_svy <- as.data.frame(qdat_strict)
  for (v in c("phenoage_accel", "mixture_qscore", "ckd", "age", "pir", "smoker", "bmi", "weight_combine
    if (v %in% names(qdat_strict_svy)) qdat_strict_svy[[v]] <- suppressWarnings(as.numeric(as.character
  }
  qdat_strict_svy$ckd <- as.integer(qdat_strict_svy$ckd)
  qdat_strict_svy$sex <- droplevels(factor(qdat_strict_svy$sex))
  qdat_strict_svy$race_eth <- droplevels(factor(qdat_strict_svy$race_eth))
```

```r
  qdat_strict_svy$cycle <- droplevels(factor(qdat_strict_svy$cycle))
  tryCatch(
    svydesign(id = ~psu, strata = ~strata, weights = ~weight_combined, nest = TRUE, data = qdat_strict_s
    error = function(e) svydesign(id = ~1, strata = NULL, weights = ~weight_combined, data = qdat_stric
  )
} else {
  NULL
}

# Model B: outcome is phenoage_accel (already age-residualized); do not adjust for age again (no double
covars_int <- c(
  if (valid_cat("sex", qdat)) "factor(sex)",
  if (valid_cat("race_eth", qdat)) "factor(race_eth)",
  if (valid_num("pir", qdat)) "pir",
  if (valid_num("smoker", qdat)) "smoker",
  if (valid_num("bmi", qdat)) "bmi",
  if (valid_cat("cycle", qdat)) "factor(cycle)"
)
intended_covars_int <- covars_int
build_int_formula <- function(covars_vec) {
  rhs <- if (length(covars_vec) > 0) paste("+", paste(covars_vec, collapse = " + ")) else ""
  as.formula(paste("phenoage_accel ~ mixture_qscore * ckd", rhs))
}
f_int <- build_int_formula(covars_int)

repeat {
  mf_int <- tryCatch(model.frame(f_int, data = qdat, na.action = na.omit), error = function(e) NULL)
  if (is.null(mf_int) || nrow(mf_int) == 0) {
    stop("Model B preprocessing failed: no complete-case rows available for interaction formula.")
  }
  factor_terms <- names(mf_int)[vapply(mf_int, is.factor, logical(1))]
  bad_factor_terms <- factor_terms[vapply(mf_int[factor_terms], function(x) nlevels(droplevels(x)) < 2,
  removable_terms <- intersect(covars_int, bad_factor_terms)
  if (length(removable_terms) == 0) break
  message("Model B: dropping collapsed factor term(s) after complete-case filtering: ",
          paste(removable_terms, collapse = ", "))
  covars_int <- setdiff(covars_int, removable_terms)
  f_int <- build_int_formula(covars_int)
}
n_modelb <- nrow(mf_int)
realized_covars_int <- covars_int
covar_report <- tibble(
  intended_term = intended_covars_int,
  realized_in_model_b = ifelse(intended_covars_int %in% realized_covars_int, "Yes", "No")
)
kable(covar_report, caption = "Model B covariate transparency: intended vs realized adjustment terms.")
```

Table 25: Model B covariate transparency: intended vs realized adjustment terms.

intended_term	realized_in_model_b
factor(sex)	Yes

intended_term	realized_in_model_b
factor(race_eth)	Yes
pir	Yes
smoker	Yes
bmi	Yes
factor(cycle)	Yes

```
fit_int_error <- NULL
fit_int <- tryCatch(
  svyglm(f_int, design = svy_q, family = gaussian()),
  error = function(e) {
    fit_int_error <<- conditionMessage(e)
    NULL
  }
)
if (is.null(fit_int)) {
  stop("Model B (svyglm interaction) failed: ", fit_int_error)
} else {
  message("Model B fit OK. Coefficients:")
  message("Model B complete-case N: ", n_modelb)
  print(coef(fit_int))
}
```

```
##                        (Intercept)                     mixture_qscore
##                        -10.2304717                        -0.7731055
##                                ckd                     factor(sex)Male
##                          9.7160990                          2.1248385
## factor(race_eth)Non-Hispanic Asian factor(race_eth)Non-Hispanic Black
##                          0.2466249                          2.4308155
## factor(race_eth)Non-Hispanic White        factor(race_eth)Other/Multi
##                          0.2626766                         -0.5525380
##                                pir                             smoker
##                         -0.5863572                          2.3821428
##                                bmi             factor(cycle)2007-2008
##                          0.3269308                         -0.9323133
##             factor(cycle)2009-2010             factor(cycle)2015-2016
##                         -0.6667737                          4.2346060
##             factor(cycle)2017-2018                 mixture_qscore:ckd
##                          5.3702929                         -1.4514657
```

```
# Sanity check: weighted lm and svyglm (weights-only design) should match.
fit_int_lm <- lm(f_int, data = qdat_svy, weights = weight_combined)
message("Weighted lm mixture_qscore coef: ", round(coef(fit_int_lm)["mixture_qscore"], 6), "; svyglm: "

aqg0 <- analytic_qg %>% filter(ckd == 0)
aqg1 <- analytic_qg %>% filter(ckd == 1)
if (nrow(aqg0) > 0) aqg0$weight_normalized <- aqg0$weight_combined / mean(aqg0$weight_combined, na.rm =
if (nrow(aqg1) > 0) aqg1$weight_normalized <- aqg1$weight_combined / mean(aqg1$weight_combined, na.rm =
fit_qg_ckd0 <- tryCatch(
  qgcomp::qgcomp.noboot(
    f_qg, expnms = qg_exposure_vars,
    data = aqg0,
```

```
    family = gaussian(), q = q_qg,
    weights = aqg0$weight_normalized
  ),
  error = function(e) NULL
)
fit_qg_ckd1 <- tryCatch(
  qgcomp::qgcomp.noboot(
    f_qg, expnms = qg_exposure_vars,
    data = aqg1,
    family = gaussian(), q = q_qg,
    weights = aqg1$weight_normalized
  ),
  error = function(e) NULL
)
```

Sample Flow (PRISMA-style)

```
n0 <- nrow(nhanes_raw)
n1 <- sum(dat$age >= config$min_age, na.rm = TRUE)
n2 <- sum(dat$age >= config$min_age & !is.na(dat$phenoage) & !is.na(dat$ckd), na.rm = TRUE)
n3 <- nrow(analytic)
n4 <- if (exists("qdat")) nrow(qdat) else NA_integer_
n5 <- if (exists("analytic_qg")) nrow(analytic_qg) else NA_integer_
n6 <- if (exists("qdat_strict")) nrow(qdat_strict) else NA_integer_
n7 <- if (exists("n_modelb")) n_modelb else NA_integer_

flow <- tibble(
  step = c(
    "Downloaded NHANES participants (selected cycles)",
    paste0("Age >= ", config$min_age),
    "Non-missing PhenoAge and CKD",
    "Non-missing selected weight, PSU, and strata (analytic sample)",
    "Primary interaction model sample (fixed 4-PFAS complete-case score + design vars)",
    "Complete-case qgcomp sample",
    "Strict complete-case (qgcomp-compatible PFAS subset)",
    "Model B complete-case estimation sample"
  ),
  included = c(n0, n1, n2, n3, n4, n5, n6, n7)
) %>%
  mutate(
    excluded_at_step = pmax(lag(included, default = first(included)) - included, 0),
    excluded_at_step = if_else(row_number() == 1, NA_integer_, as.integer(excluded_at_step)),
    excluded_label = ifelse(is.na(excluded_at_step), "", paste0("Excluded: ", excluded_at_step))
  )

kable(flow, caption = "Sample flow counts (PRISMA-style): inclusion and exclusion at each analytic step
```

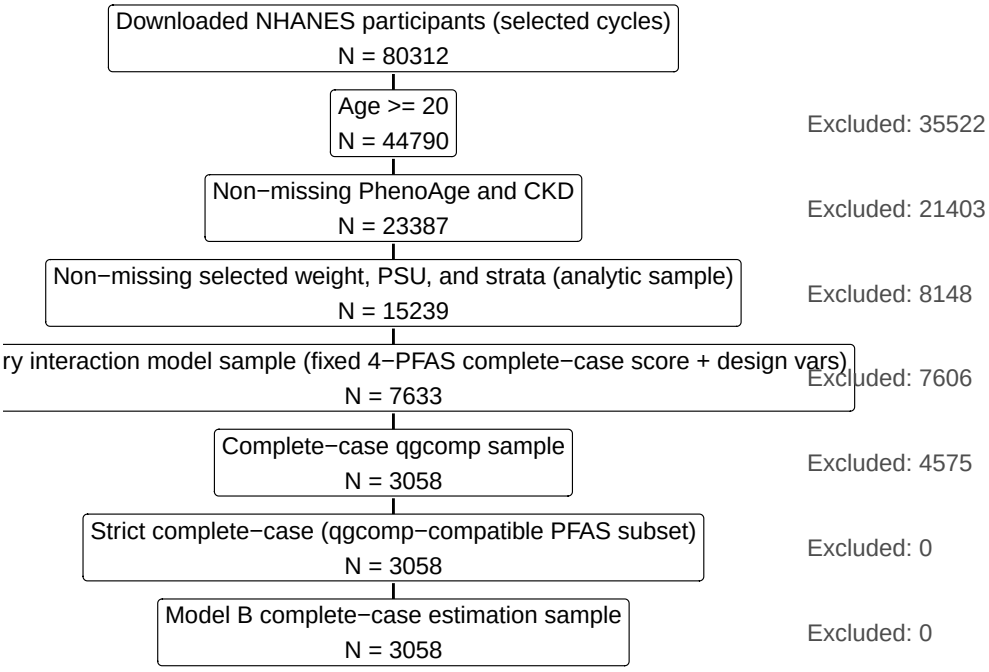
Table 26: Sample flow counts (PRISMA-style): inclusion and exclusion at each analytic step

step	included	excluded_at_step	excluded_label
Downloaded NHANES participants (selected cycles)	80312	NA	
Age >= 20	44790	35522	Excluded: 35522
Non-missing PhenoAge and CKD	23387	21403	Excluded: 21403
Non-missing selected weight, PSU, and strata (analytic sample)	15239	8148	Excluded: 8148
Primary interaction model sample (fixed 4-PFAS complete-case score + design vars)	7633	7606	Excluded: 7606
Complete-case qgcomp sample	3058	4575	Excluded: 4575
Strict complete-case (qgcomp-compatible PFAS subset)	3058	0	Excluded: 0
Model B complete-case estimation sample	3058	0	Excluded: 0

```r
flow_plot <- flow %>%
  mutate(
    y = rev(seq_len(n())),
    box_label = paste0(step, "\nN = ", included)
  )

ggplot(flow_plot, aes(x = 1, y = y)) +
  geom_segment(
    data = flow_plot %>% filter(y > min(y)),
    aes(x = 1, xend = 1, y = y - 0.42, yend = y - 0.95),
    arrow = arrow(length = unit(0.12, "inches")),
    linewidth = 0.5
  ) +
  geom_label(aes(label = box_label), size = 3.2, label.size = 0.2, fill = "white") +
  geom_text(aes(x = 1.62, label = excluded_label), hjust = 0, size = 3.1, color = "gray30") +
  xlim(0.5, 2.2) +
  ylim(0.5, max(flow_plot$y) + 0.8) +
  labs(
    title = "Participant flow from downloaded sample to analytic samples",
    subtitle = "PRISMA-style flow for NHANES analytic cohort development",
    x = NULL, y = NULL
  ) +
  theme_void(base_size = 11)
```

Participant flow from downloaded sample to analytic samples

PRISMA−style flow for NHANES analytic cohort development



Tables

```
fmt <- function(x, digits = 3) {
  ifelse(
    is.na(x),
    "NA",
    ifelse(
      x != 0 & abs(x) < 10^-digits,
      format(x, scientific = TRUE, digits = 2),
      format(round(x, digits), nsmall = digits)
    )
  )
}
fmt_p <- function(x) ifelse(is.na(x), "NA", ifelse(x < 0.001, "<0.001", format(round(x, 3), nsmall = 3)

# Table 1: characteristics by CKD -- continuous: mean (SD); categorical: n (weighted %)
t1_mean_sd <- function(formula, design) {
  est <- svymean(formula, design, na.rm = TRUE)
  v_obj <- svyvar(formula, design, na.rm = TRUE)
  sd_val <- sqrt(as.numeric(coef(v_obj)[1]))
  paste0(format(round(coef(est)[1], 2), nsmall = 2), " (", format(round(sd_val, 2), nsmall = 2), ")")
}
t1_mean_obj <- function(formula, design) {
```

```r
  m <- svymean(formula, design, na.rm = TRUE)
  v <- svyvar(formula, design, na.rm = TRUE)
  list(est = unname(coef(m)[1]), se = unname(SE(m)[1]), var = unname(coef(v)[1]))
}
t1_pct_se <- function(formula, design) {
  est <- svymean(formula, design, na.rm = TRUE)
  cf <- coef(est)
  se <- SE(est)
  idx <- 1
  if (length(cf) > 1) {
    nms <- names(cf)
    idx_true <- which(str_detect(nms, "TRUE$"))
    if (length(idx_true) == 0) idx_true <- which(str_detect(nms, "1$"))
    if (length(idx_true) > 0) idx <- idx_true[1]
  }
  paste0(
    format(round(unname(cf[idx]) * 100, 1), nsmall = 1),
    " (",
    format(round(unname(se[idx]) * 100, 1), nsmall = 1),
    ")"
  )
}
t1_n_wpct <- function(cond_vec, formula, design) {
  n_unwt <- sum(cond_vec, na.rm = TRUE)
  est <- svymean(formula, design, na.rm = TRUE)
  cf <- coef(est)
  idx <- 1
  if (length(cf) > 1) {
    nms <- names(cf)
    idx_true <- which(str_detect(nms, "TRUE$"))
    if (length(idx_true) == 0) idx_true <- which(str_detect(nms, "1$"))
    if (length(idx_true) > 0) idx <- idx_true[1]
  }
  paste0(
    format(n_unwt, big.mark = ","),
    " (",
    format(round(unname(cf[idx]) * 100, 1), nsmall = 1),
    ")"
  )
}
t1_pct_obj <- function(formula, design) {
  est <- svymean(formula, design, na.rm = TRUE)
  cf <- coef(est)
  se <- SE(est)
  idx <- 1
  if (length(cf) > 1) {
    nms <- names(cf)
    idx_true <- which(str_detect(nms, "TRUE$"))
    if (length(idx_true) == 0) idx_true <- which(str_detect(nms, "1$"))
    if (length(idx_true) > 0) idx <- idx_true[1]
  }
  list(est = unname(cf[idx]), se = unname(se[idx]))
}
```

```r
t1_factor_level_pct_se <- function(var, level, design) {
  est <- svymean(as.formula(paste0("~factor(", var, ")")), design, na.rm = TRUE)
  key <- paste0("factor(", var, ")", level)
  if (!(key %in% names(coef(est)))) return("NA")
  paste0(
    format(round(unname(coef(est)[key]) * 100, 1), nsmall = 1),
    " (",
    format(round(unname(SE(est)[key]) * 100, 1), nsmall = 1),
    ")"
  )
}
t1_factor_level_n_wpct <- function(var, level, design, data) {
  n_unwt <- sum(data[[var]] == level, na.rm = TRUE)
  est <- svymean(as.formula(paste0("~factor(", var, ")")), design, na.rm = TRUE)
  key <- paste0("factor(", var, ")", level)
  if (!(key %in% names(coef(est)))) {
    return(paste0(format(n_unwt, big.mark = ","), " (NA)"))
  }
  paste0(
    format(n_unwt, big.mark = ","),
    " (",
    format(round(unname(coef(est)[key]) * 100, 1), nsmall = 1),
    ")"
  )
}
t1_factor_level_obj <- function(var, level, design) {
  est <- svymean(as.formula(paste0("~factor(", var, ")")), design, na.rm = TRUE)
  key <- paste0("factor(", var, ")", level)
  if (!(key %in% names(coef(est)))) return(list(est = NA_real_, se = NA_real_))
  list(est = unname(coef(est)[key]), se = unname(SE(est)[key]))
}
smd_cont <- function(o0, o1) {
  if (any(!is.finite(c(o0$est, o1$est, o0$var, o1$var)))) return(NA_real_)
  denom <- sqrt((o0$var + o1$var) / 2)
  if (!is.finite(denom) || denom <= 0) return(NA_real_)
  abs(o1$est - o0$est) / denom
}
smd_prop <- function(o0, o1) {
  if (any(!is.finite(c(o0$est, o1$est)))) return(NA_real_)
  denom <- sqrt((o0$est * (1 - o0$est) + o1$est * (1 - o1$est)) / 2)
  if (!is.finite(denom) || denom <= 0) return(NA_real_)
  abs(o1$est - o0$est) / denom
}
fmt_smd <- function(x) ifelse(is.na(x), "", format(round(x, 3), nsmall = 3))

svy0 <- subset(svy, ckd == 0)
svy1 <- subset(svy, ckd == 1)
analytic0 <- analytic[!is.na(analytic$ckd) & analytic$ckd == 0, ]
analytic1 <- analytic[!is.na(analytic$ckd) & analytic$ckd == 1, ]
age0 <- t1_mean_obj(~age, svy0); age1 <- t1_mean_obj(~age, svy1)
fem0 <- t1_pct_obj(~I(sex == "Female"), svy0); fem1 <- t1_pct_obj(~I(sex == "Female"), svy1)
his0 <- t1_factor_level_obj("race_eth", "Hispanic", svy0); his1 <- t1_factor_level_obj("race_eth", "His
whi0 <- t1_factor_level_obj("race_eth", "Non-Hispanic White", svy0); whi1 <- t1_factor_level_obj("race_
```

```
blk0 <- t1_factor_level_obj("race_eth", "Non-Hispanic Black", svy0); blk1 <- t1_factor_level_obj("race_
asn0 <- t1_factor_level_obj("race_eth", "Non-Hispanic Asian", svy0); asn1 <- t1_factor_level_obj("race_
oth0 <- t1_factor_level_obj("race_eth", "Other/Multi", svy0); oth1 <- t1_factor_level_obj("race_eth", "
edu_lths0 <- t1_factor_level_obj("education", "<HS", svy0); edu_lths1 <- t1_factor_level_obj("education
edu_hs0 <- t1_factor_level_obj("education", "HS/GED", svy0); edu_hs1 <- t1_factor_level_obj("education"
edu_gt0 <- t1_factor_level_obj("education", ">HS", svy0); edu_gt1 <- t1_factor_level_obj("education", "
pir0 <- t1_mean_obj(~pir, svy0); pir1 <- t1_mean_obj(~pir, svy1)
bmi0 <- t1_mean_obj(~bmi, svy0); bmi1 <- t1_mean_obj(~bmi, svy1)
smk0 <- t1_pct_obj(~I(smoker == 1), svy0); smk1 <- t1_pct_obj(~I(smoker == 1), svy1)
alc0 <- t1_pct_obj(~I(alcohol == 1), svy0); alc1 <- t1_pct_obj(~I(alcohol == 1), svy1)
pa0 <- t1_pct_obj(~I(phys_active == 1), svy0); pa1 <- t1_pct_obj(~I(phys_active == 1), svy1)
dm0 <- t1_pct_obj(~I(diabetes == 1), svy0); dm1 <- t1_pct_obj(~I(diabetes == 1), svy1)
htn0 <- t1_pct_obj(~I(hypertension == 1), svy0); htn1 <- t1_pct_obj(~I(hypertension == 1), svy1)
cvd0 <- t1_pct_obj(~I(cvd == 1), svy0); cvd1 <- t1_pct_obj(~I(cvd == 1), svy1)

table1 <- tibble(
  characteristic = c(
    "N (unweighted)", "Age, mean (SD)", "Female, n (weighted %)",
    "Hispanic, n (weighted %)", "Non-Hispanic White, n (weighted %)", "Non-Hispanic Black, n (weighted 
    "Non-Hispanic Asian, n (weighted %)", "Other/Multi, n (weighted %)",
    "Education <HS, n (weighted %)", "Education HS/GED, n (weighted %)", "Education >HS, n (weighted %)
    "PIR, mean (SD)", "BMI, mean (SD)",
    "Current smoker, n (weighted %)", "Alcohol (yes), n (weighted %)", "Physically active, n (weighted 
    "Diabetes, n (weighted %)", "Hypertension, n (weighted %)", "CVD history, n (weighted %)"
  ),
  `No CKD` = c(
    as.character(sum(analytic$ckd == 0, na.rm = TRUE)),
    t1_mean_sd(~age, svy0),
    t1_n_wpct(analytic0$sex == "Female", ~I(sex == "Female"), svy0),
    t1_factor_level_n_wpct("race_eth", "Hispanic", svy0, analytic0),
    t1_factor_level_n_wpct("race_eth", "Non-Hispanic White", svy0, analytic0),
    t1_factor_level_n_wpct("race_eth", "Non-Hispanic Black", svy0, analytic0),
    t1_factor_level_n_wpct("race_eth", "Non-Hispanic Asian", svy0, analytic0),
    t1_factor_level_n_wpct("race_eth", "Other/Multi", svy0, analytic0),
    t1_factor_level_n_wpct("education", "<HS", svy0, analytic0),
    t1_factor_level_n_wpct("education", "HS/GED", svy0, analytic0),
    t1_factor_level_n_wpct("education", ">HS", svy0, analytic0),
    t1_mean_sd(~pir, svy0),
    t1_mean_sd(~bmi, svy0),
    t1_n_wpct(analytic0$smoker == 1, ~I(smoker == 1), svy0),
    t1_n_wpct(analytic0$alcohol == 1, ~I(alcohol == 1), svy0),
    t1_n_wpct(analytic0$phys_active == 1, ~I(phys_active == 1), svy0),
    t1_n_wpct(analytic0$diabetes == 1, ~I(diabetes == 1), svy0),
    t1_n_wpct(analytic0$hypertension == 1, ~I(hypertension == 1), svy0),
    t1_n_wpct(analytic0$cvd == 1, ~I(cvd == 1), svy0)
  ),
  CKD = c(
    as.character(sum(analytic$ckd == 1, na.rm = TRUE)),
    t1_mean_sd(~age, svy1),
    t1_n_wpct(analytic1$sex == "Female", ~I(sex == "Female"), svy1),
    t1_factor_level_n_wpct("race_eth", "Hispanic", svy1, analytic1),
    t1_factor_level_n_wpct("race_eth", "Non-Hispanic White", svy1, analytic1),
    t1_factor_level_n_wpct("race_eth", "Non-Hispanic Black", svy1, analytic1),
```

```r
    t1_factor_level_n_wpct("race_eth", "Non-Hispanic Asian", svy1, analytic1),
    t1_factor_level_n_wpct("race_eth", "Other/Multi", svy1, analytic1),
    t1_factor_level_n_wpct("education", "<HS", svy1, analytic1),
    t1_factor_level_n_wpct("education", "HS/GED", svy1, analytic1),
    t1_factor_level_n_wpct("education", ">HS", svy1, analytic1),
    t1_mean_sd(~pir, svy1),
    t1_mean_sd(~bmi, svy1),
    t1_n_wpct(analytic1$smoker == 1, ~I(smoker == 1), svy1),
    t1_n_wpct(analytic1$alcohol == 1, ~I(alcohol == 1), svy1),
    t1_n_wpct(analytic1$phys_active == 1, ~I(phys_active == 1), svy1),
    t1_n_wpct(analytic1$diabetes == 1, ~I(diabetes == 1), svy1),
    t1_n_wpct(analytic1$hypertension == 1, ~I(hypertension == 1), svy1),
    t1_n_wpct(analytic1$cvd == 1, ~I(cvd == 1), svy1)
  ),
  SMD = c(
    "",
    fmt_smd(smd_cont(age0, age1)),
    fmt_smd(smd_prop(fem0, fem1)),
    fmt_smd(smd_prop(his0, his1)),
    fmt_smd(smd_prop(whi0, whi1)),
    fmt_smd(smd_prop(blk0, blk1)),
    fmt_smd(smd_prop(asn0, asn1)),
    fmt_smd(smd_prop(oth0, oth1)),
    fmt_smd(smd_prop(edu_lths0, edu_lths1)),
    fmt_smd(smd_prop(edu_hs0, edu_hs1)),
    fmt_smd(smd_prop(edu_gt0, edu_gt1)),
    fmt_smd(smd_cont(pir0, pir1)),
    fmt_smd(smd_cont(bmi0, bmi1)),
    fmt_smd(smd_prop(smk0, smk1)),
    fmt_smd(smd_prop(alc0, alc1)),
    fmt_smd(smd_prop(pa0, pa1)),
    fmt_smd(smd_prop(dm0, dm1)),
    fmt_smd(smd_prop(htn0, htn1)),
    fmt_smd(smd_prop(cvd0, cvd1))
  )
)

kable(table1, caption = "Table 1. Sample characteristics by CKD status [continuous: mean (SD); categori
```

Table 27: Table 1. Sample characteristics by CKD status [continuous: mean (SD); categorical: n (weighted %)] with SMD

characteristic	No CKD	CKD	SMD
N (unweighted)	12563	2676	
Age, mean (SD)	45.21 (15.85)	60.71 (17.18)	0.938
Female, n (weighted %)	6,513 (51.2)	1,397 (57.4)	0.126
Hispanic, n (weighted %)	3,554 (14.4)	620 (12.9)	0.043
Non-Hispanic White, n (weighted %)	5,337 (67.3)	1,194 (65.8)	0.033
Non-Hispanic Black, n (weighted %)	2,409 (10.4)	662 (14.3)	0.120
Non-Hispanic Asian, n (weighted %)	672 (2.5)	93 (2.1)	0.029
Other/Multi, n (weighted %)	591 (5.5)	107 (5.0)	0.022
Education <HS, n (weighted %)	1,338 (5.8)	454 (12.2)	0.225

characteristic	No CKD	CKD	SMD
Education HS/GED, n (weighted %)	2,857 (26.1)	676 (32.7)	0.145
Education >HS, n (weighted %)	6,578 (68.1)	1,089 (55.1)	0.269
PIR, mean (SD)	3.10 (1.63)	2.62 (1.56)	0.301
BMI, mean (SD)	28.77 (6.76)	30.43 (7.85)	0.227
Current smoker, n (weighted %)	2,663 (47.1)	443 (31.8)	0.317
Alcohol (yes), n (weighted %)	636 (15.4)	150 (16.1)	0.019
Physically active, n (weighted %)	4,980 (55.4)	720 (40.7)	0.297
Diabetes, n (weighted %)	1,131 (6.7)	888 (28.3)	0.593
Hypertension, n (weighted %)	3,735 (26.7)	1,717 (60.1)	0.715
CVD history, n (weighted %)	891 (5.7)	724 (23.5)	0.522

```
cat("\n*Table 1 note:* Weighted estimates use NHANES complex survey design (strata, PSU, combined weigh
    "SMD is the absolute standardized mean difference between CKD and No CKD groups; values >0.10 are t
```

```
##
## *Table 1 note:* Weighted estimates use NHANES complex survey design (strata, PSU, combined weights).
```

```
# Table 2: overall mixture effect
tab2 <- safe_qgcomp_extract(
  fit_qg_overall,
  "Model A: Overall mixture effect",
  nrow(analytic_qg)
) %>%
  mutate(
    beta = fmt(beta),
    conf.low = fmt(conf.low),
    conf.high = fmt(conf.high),
    p.value = fmt_p(p.value),
    interpretation = "Change in PhenoAge acceleration (years)",
    note = if (is.null(fit_qg_overall)) {
      ifelse(is.null(qgcomp_overall_error), "qgcomp failed (unknown error)", qgcomp_overall_error)
    } else {
      "OK"
    }
  )
kable(
  tab2,
  caption = paste0(
    "Table 2. Descriptive overall association between environmental mixture score and ",
    "PhenoAge acceleration (years). ",
    "Mixture score ranges from 1 to 4; a 1-unit increase represents a 1-quartile increase averaged ",
    "across the fixed primary PFAS set. Estimates from qgcomp with normalized pooled NHANES sampling we
  )
)
```

Table 28: Table 2. Descriptive overall association between environmental mixture score and PhenoAge acceleration (years). Mixture score ranges from 1 to 4; a 1-unit increase represents a 1-quartile increase averaged across the fixed primary PFAS set. Estimates from qgcomp with normalized pooled NHANES sampling weights.

model	beta	conf.low	conf.high	p.value	N	interpretation	note
Model A: Overall mixture effect	-1.195	-1.540	-0.850	<0.001	3058	Change in PhenoAge acceleration (years)	OK

```
cat("\n*Table 2 note:* qgcomp model adjusted for same 6 variables as Model B: sex, race/ethnicity, PIR,
    "qgcomp intended covariates: ", paste(qg_covars_intended, collapse = ", "), ". ",
    "qgcomp realized covariates (after complete-case support checks): ", paste(qg_covars_realized, colla
    "This table is descriptive/directional; primary inference is based on Table 3 survey-weighted intera
    "PhenoAge acceleration defined as survey-weighted residual of PhenoAge on chronological age. ",
    "N reflects unweighted analytic sample used in mixture modeling.\n", sep = "")
```

```
##
## *Table 2 note:* qgcomp model adjusted for same 6 variables as Model B: sex, race/ethnicity, PIR, BMI
```

```
# Table 3: interaction results + design-based marginal effects by CKD group
if (!is.null(fit_int)) {
  int_tidy <- broom::tidy(fit_int, conf.int = TRUE)
  cf_names <- names(coef(fit_int))

  int_terms_wanted <- c("mixture_qscore", cf_names[str_detect(cf_names, "^ckd$|mixture_qscore:ckd")])
  tab3_raw <- int_tidy %>%
    filter(term %in% int_terms_wanted) %>%
    transmute(
      term,
      beta = fmt(estimate),
      conf.low = fmt(conf.low),
      conf.high = fmt(conf.high),
      p.value = fmt_p(p.value),
      N = n_modelb,
      model = "Model B: Mixture*CKD interaction"
    )

  # Design-based marginal effects via svycontrast with named contrast vectors.

  # Robustly identify the interaction term regardless of factor coding.
  int_name <- cf_names[str_detect(cf_names, "mixture_qscore:ckd")]
  if (length(int_name) != 1) {
    warning("Could not uniquely identify mixture*ckd interaction term among: ",
            paste(cf_names, collapse = ", "), ". Skipping marginal effects.")
    tab3 <- tab3_raw
  } else {
    c_ckd0 <- setNames(rep(0, length(cf_names)), cf_names)
    c_ckd0["mixture_qscore"] <- 1

    c_ckd1 <- setNames(rep(0, length(cf_names)), cf_names)
    c_ckd1["mixture_qscore"] <- 1
```

```r
    c_ckd1[int_name] <- 1

    contrast_ckd0 <- svycontrast(fit_int, c_ckd0)
    contrast_ckd1 <- svycontrast(fit_int, c_ckd1)

    ddf <- fit_int$df.residual
    if (is.null(ddf) || !is.finite(ddf)) ddf <- Inf

    make_contrast_row <- function(ctr, label, df) {
      est <- as.numeric(coef(ctr))
      se  <- as.numeric(SE(ctr))
      t_val <- est / se
      p_val <- 2 * pt(-abs(t_val), df = df)
      tibble(term = label, beta = fmt(est),
             conf.low = fmt(est - qt(0.975, df) * se),
             conf.high = fmt(est + qt(0.975, df) * se),
             p.value = fmt_p(p_val), N = n_modelb, model = "Marginal effects (design-based)")
    }

    p_int_tidy <- int_tidy %>% filter(term == int_name) %>% pull(p.value)
    p_interaction <- if (length(p_int_tidy) > 0) p_int_tidy[1] else NA_real_

    tab3_marginal <- bind_rows(
      make_contrast_row(contrast_ckd0, "Mixture effect among CKD=0", ddf),
      make_contrast_row(contrast_ckd1, "Mixture effect among CKD=1", ddf),
      tibble(term = "P-interaction", beta = "", conf.low = "", conf.high = "",
             p.value = fmt_p(p_interaction), N = n_modelb, model = "Marginal effects (design-based)")
    )
    tab3 <- bind_rows(tab3_raw, tab3_marginal)
  }
} else {
  tab3 <- tibble(
    term = c("mixture_qscore", "ckd", "mixture_qscore:ckd"),
    beta = "NA", conf.low = "NA", conf.high = "NA", p.value = "NA",
    N = n_modelb, model = "Model B: Mixture*CKD interaction (not estimable)"
  )
}
kable(tab3, caption = "Table 3. Interaction model with design-based marginal effects by CKD group")
```

Table 29: Table 3. Interaction model with design-based marginal effects by CKD group

term	beta	conf.low	conf.high	p.value	N	model
mixture_qscore	-0.773	-1.077	-0.469	<0.001	3058	Model B: Mixture*CKD interaction
ckd	9.716	6.255	13.177	<0.001	3058	Model B: Mixture*CKD interaction
mixture_qscore:ckd	-1.451	-2.615	-0.288	0.015	3058	Model B: Mixture*CKD interaction
Mixture effect among CKD=0	-0.773	-1.077	-0.469	<0.001	3058	Marginal effects (design-based)

term	beta	conf.low	conf.high	p.value	N	model
Mixture effect among CKD=1	-2.225	-3.342	-1.107	<0.001	3058	Marginal effects (design-based)
P-interaction				0.015	3058	Marginal effects (design-based)

```
cat("\n*Table 3 note:* Model B is a survey-weighted linear model (`svyglm`) with interaction term `mixtu
    "Intended covariates: ", paste(intended_covars_int, collapse = ", "), ". ",
    "Realized covariates (after complete-case support checks): ", paste(realized_covars_int, collapse =
    "Marginal effects for CKD=0 and CKD=1 are estimated using design-based contrasts (`svycontrast`) wi
```

```
##
## *Table 3 note:* Model B is a survey-weighted linear model (‘svyglm‘) with interaction term ‘mixture_c
```

```
# Table 4: sensitivity analyses -- using same covariate set as primary models.
sens_covars_rhs <- paste(realized_covars_int, collapse = " + ")
make_rhs_formula <- function(outcome, main_terms, covar_rhs) {
  rhs <- paste(main_terms, collapse = " + ")
  if (!is.null(covar_rhs) && nzchar(covar_rhs)) rhs <- paste(rhs, covar_rhs, sep = " + ")
  as.formula(paste(outcome, "~", rhs))
}
fit_svy_with_covar_guard <- function(data_df, outcome, main_terms, covars_terms, label) {
  build_formula <- function(covs) {
    covar_rhs <- if (length(covs) > 0) paste(covs, collapse = " + ") else ""
    make_rhs_formula(outcome, main_terms, covar_rhs)
  }
  covs <- covars_terms
  intended <- covs
  f <- build_formula(covs)
  mf <- NULL
  repeat {
    mf <- tryCatch(model.frame(f, data = data_df, na.action = na.omit), error = function(e) NULL)
    if (is.null(mf) || nrow(mf) == 0) break
    factor_terms <- names(mf)[vapply(mf, is.factor, logical(1))]
    bad_factor_terms <- factor_terms[vapply(mf[factor_terms], function(x) nlevels(droplevels(x)) < 2, le
    removable <- intersect(covs, bad_factor_terms)
    if (length(removable) == 0) break
    message("Sensitivity [", label, "]: dropping collapsed factor term(s): ", paste(removable, collapse
    covs <- setdiff(covs, removable)
    f <- build_formula(covs)
  }
  if (is.null(mf) || nrow(mf) == 0) {
    return(list(
      fit = NULL, n_model = 0L, intended = intended, realized = covs,
      dropped = paste(setdiff(intended, covs), collapse = ", "),
      error = "No complete-case rows for model frame."
    ))
  }
  data_svy <- as.data.frame(data_df)
  for (v in c("phenoage_accel", "phenoage_nokid", "phenoage_nokid_both", "phenoage_accel_unweighted", "n
    if (v %in% names(data_svy)) data_svy[[v]] <- suppressWarnings(as.numeric(as.character(data_svy[[v]]))
  }
```

```r
  if ("ckd" %in% names(data_svy)) data_svy$ckd <- as.integer(data_svy$ckd)
  for (v in c("sex", "race_eth", "cycle")) {
    if (v %in% names(data_svy)) data_svy[[v]] <- factor(as.character(data_svy[[v]]))
  }
  data_svy$sex      <- droplevels(factor(data_svy$sex))
  data_svy$race_eth <- droplevels(factor(data_svy$race_eth))
  data_svy$cycle    <- droplevels(factor(data_svy$cycle))
  des <- tryCatch(
    svydesign(id = ~psu, strata = ~strata, weights = ~weight_combined, nest = TRUE, data = data_svy),
    error = function(e) svydesign(id = ~1, strata = NULL, weights = ~weight_combined, data = data_svy)
  )
  err <- NULL
  fit <- tryCatch(
    svyglm(f, design = des, family = gaussian()),
    error = function(e) {
      err <<- conditionMessage(e)
      NULL
    }
  )
  list(
    fit = fit,
    n_model = nrow(mf),
    intended = intended,
    realized = covs,
    dropped = paste(setdiff(intended, covs), collapse = ", "),
    error = err
  )
}

sens_ckd_egfr <- qdat %>%
  mutate(ckd_sens = ifelse(!is.na(egfr), as.integer(egfr < 60), NA_integer_)) %>%
  filter(!is.na(ckd_sens))
fit_sens1_obj <- fit_svy_with_covar_guard(sens_ckd_egfr, "phenoage_accel", c("mixture_qscore", "ckd_sens
fit_sens1 <- fit_sens1_obj$fit

sens_pheno <- qdat
ph2_comp <- compute_phenoage_components(sens_pheno, include_scr = FALSE, include_alb = TRUE)
ph2 <- ph2_comp$phenoage
sens_pheno$phenoage_nokid <- NA_real_
keep2 <- which(!is.na(ph2) & !is.na(sens_pheno$age))
if (length(keep2) > 0) {
  fit_nokid <- lm(ph2[keep2] ~ sens_pheno$age[keep2])
  sens_pheno$phenoage_nokid[keep2] <- residuals(fit_nokid)
}
fit_sens2_obj <- fit_svy_with_covar_guard(sens_pheno, "phenoage_nokid", c("mixture_qscore", "ckd"), real
fit_sens2 <- fit_sens2_obj$fit

ph3_comp <- compute_phenoage_components(sens_pheno, include_scr = FALSE, include_alb = FALSE)
ph3 <- ph3_comp$phenoage
sens_pheno$phenoage_nokid_both <- NA_real_
keep3 <- which(!is.na(ph3) & !is.na(sens_pheno$age))
if (length(keep3) > 0) {
  fit_nokid_both <- lm(ph3[keep3] ~ sens_pheno$age[keep3])
```

```r
  sens_pheno$phenoage_nokid_both[keep3] <- residuals(fit_nokid_both)
}
fit_sens3_obj <- fit_svy_with_covar_guard(sens_pheno, "phenoage_nokid_both", c("mixture_qscore", "ckd")
fit_sens3 <- fit_sens3_obj$fit

get_mix_row <- function(fit, label, n_model = NA_integer_) {
  if (is.null(fit)) return(tibble(sensitivity = label, estimate = NA_real_, conf.low = NA_real_, conf.hi
  tb <- broom::tidy(fit, conf.int = TRUE) %>% filter(term == "mixture_qscore")
  if (nrow(tb) == 0) return(tibble(sensitivity = label, estimate = NA_real_, conf.low = NA_real_, conf.h
  tb %>% mutate(sensitivity = label, N = n_model) %>% select(sensitivity, estimate, conf.low, conf.high
}

f_sens_uw <- make_rhs_formula("phenoage_accel_unweighted", c("mixture_qscore", "ckd"), sens_covars_rhs)
f_sens_both <- make_rhs_formula("phenoage_accel", c("mixture_qscore", "ckd"), sens_covars_rhs)
fit_sens_uw_obj <- fit_svy_with_covar_guard(qdat, "phenoage_accel_unweighted", c("mixture_qscore", "ckd
fit_sens_both_markers_obj <- fit_svy_with_covar_guard(qdat %>% filter(!is.na(egfr), !is.na(acr)), "phen
fit_sens_no_acute_obj <- fit_svy_with_covar_guard(qdat %>% filter(is.na(crp_acute) | crp_acute == 0L),
fit_sens_strict_obj <- fit_svy_with_covar_guard(qdat_strict, "phenoage_accel", c("mixture_qscore", "ckd

# Sensitivity: additional adjustment for cardiometabolic conditions potentially on the causal pathway.
covars_cardiometabolic <- c(
  realized_covars_int,
  if (valid_num("diabetes", qdat)) "diabetes",
  if (valid_num("hypertension", qdat)) "hypertension",
  if (valid_num("cvd", qdat)) "cvd"
)
fit_sens_cardiometabolic_obj <- fit_svy_with_covar_guard(qdat, "phenoage_accel", c("mixture_qscore", "c
fit_sens_cardiometabolic <- fit_sens_cardiometabolic_obj$fit

tab4 <- bind_rows(
  get_mix_row(fit_sens1, "CKD redefined as eGFR<60", fit_sens1_obj$n_model),
  get_mix_row(fit_sens2, "PhenoAge excluding creatinine only", fit_sens2_obj$n_model),
  get_mix_row(fit_sens3, "PhenoAge excluding creatinine and albumin", fit_sens3_obj$n_model),
  get_mix_row(
    fit_sens_uw_obj$fit,
    "Unweighted residualized PhenoAgeAccel (sensitivity)",
    fit_sens_uw_obj$n_model
  ),
  get_mix_row(
    fit_sens_both_markers_obj$fit,
    "Restricted to participants with both eGFR and ACR",
    fit_sens_both_markers_obj$n_model
  ),
  get_mix_row(
    fit_sens_no_acute_obj$fit,
    "Exclude participants with CRP > 1 mg/dL",
    fit_sens_no_acute_obj$n_model
  ),
  get_mix_row(
    fit_sens_cardiometabolic,
    "Additional adjustment for diabetes, hypertension, CVD (potentially on causal pathway)",
    fit_sens_cardiometabolic_obj$n_model
  ),
```

```
  get_mix_row(
    fit_sens_strict_obj$fit,
    "Strict complete-case (qgcomp-compatible PFAS subset), same 6-variable adjustment",
    fit_sens_strict_obj$n_model
  )
) %>%
  transmute(
    sensitivity,
    beta = fmt(estimate),
    conf.low = fmt(conf.low),
    conf.high = fmt(conf.high),
    p.value = fmt_p(p.value),
    N,
    model = "Sensitivity"
  )
kable(tab4, caption = "Table 4. Sensitivity analyses (primary adjustment: sex, race, PIR, BMI, smoking,
```

Table 30: Table 4. Sensitivity analyses (primary adjustment: sex, race, PIR, BMI, smoking, cycle)

sensitivity	beta	conf.low	conf.high	p.value	N	model
CKD redefined as eGFR<60	-1.094	-1.385	-0.804	<0.001	3058	Sensitivity
PhenoAge excluding creatinine only	-0.869	-1.144	-0.593	<0.001	3058	Sensitivity
PhenoAge excluding creatinine and albumin	-0.698	-0.960	-0.437	<0.001	3058	Sensitivity
Unweighted residualized PhenoAgeAccel (sensitivity)	-1.048	-1.353	-0.743	<0.001	3058	Sensitivity
Restricted to participants with both eGFR and ACR	-0.947	-1.253	-0.640	<0.001	3013	Sensitivity
Exclude participants with CRP > 1 mg/dL	-1.049	-1.354	-0.744	<0.001	2952	Sensitivity
Additional adjustment for diabetes, hypertension, CVD (potentially on causal pathway)	-1.030	-1.344	-0.717	<0.001	2960	Sensitivity
Strict complete-case (qgcomp-compatible PFAS subset), same 6-variable adjustment	-1.021	-1.312	-0.729	<0.001	3058	Sensitivity

```
cat("\n*Table 4 note:* Primary and most sensitivity models use the lean adjustment set (sex, race/ethni
    "One row adds diabetes, hypertension, and CVD with explicit labeling that these may be on the causal
    "Each row reports the `mixture_qscore` coefficient from a survey-weighted model (`svyglm`) with NHA
```

```
##
## *Table 4 note:* Primary and most sensitivity models use the lean adjustment set (sex, race/ethnicity
```

```
sens_diagnostics <- bind_rows(
  tibble(sensitivity = "CKD redefined as eGFR<60", N_model = fit_sens1_obj$n_model, dropped_covariates =
  tibble(sensitivity = "PhenoAge excluding creatinine only", N_model = fit_sens2_obj$n_model, dropped_co
  tibble(sensitivity = "PhenoAge excluding creatinine and albumin", N_model = fit_sens3_obj$n_model, dro
  tibble(sensitivity = "Unweighted residualized PhenoAgeAccel (sensitivity)", N_model = fit_sens_uw_obj$
  tibble(sensitivity = "Restricted to participants with both eGFR and ACR", N_model = fit_sens_both_mar
```

```
  tibble(sensitivity = "Exclude participants with CRP > 1 mg/dL", N_model = fit_sens_no_acute_obj$n_mode
  tibble(sensitivity = "Additional adjustment for diabetes, hypertension, CVD (potentially on causal pa
  tibble(sensitivity = "Strict complete-case (qgcomp-compatible PFAS subset), same 6-variable adjustmen
)
kable(sens_diagnostics, caption = "Sensitivity model diagnostics (dropped covariates and fit errors).")
```

Table 31: Sensitivity model diagnostics (dropped covariates and fit errors).

sensitivity	N_model	dropped_covariates	fit_error
CKD redefined as eGFR<60	3058	None	None
PhenoAge excluding creatinine only	3058	None	None
PhenoAge excluding creatinine and albumin	3058	None	None
Unweighted residualized PhenoAgeAccel (sensitivity)	3058	None	None
Restricted to participants with both eGFR and ACR	3013	None	None
Exclude participants with CRP > 1 mg/dL	2952	None	None
Additional adjustment for diabetes, hypertension, CVD (potentially on causal pathway)	2960	None	None
Strict complete-case (qgcomp-compatible PFAS subset), same 6-variable adjustment	3058	None	None

```
# Supplementary: single-pollutant models with same 6-variable adjustment as Model B.
single_pollutant_out <- lapply(exposure_vars, function(v) {
  sp_obj <- fit_svy_with_covar_guard(qdat, "phenoage_accel", c(v, "ckd"), realized_covars_int, paste0("
  fit_sp <- sp_obj$fit
  if (is.null(fit_sp)) {
    return(list(
      row = tibble(exposure = v, beta = NA_real_, conf.low = NA_real_, conf.high = NA_real_, p.value = N
      diag = tibble(exposure = v, N_model = sp_obj$n_model, dropped_covariates = ifelse(sp_obj$dropped =
    ))
  }
  tb <- broom::tidy(fit_sp, conf.int = TRUE) %>% filter(term == v)
  if (nrow(tb) == 0) {
    return(list(
      row = tibble(exposure = v, beta = NA_real_, conf.low = NA_real_, conf.high = NA_real_, p.value = N
      diag = tibble(exposure = v, N_model = sp_obj$n_model, dropped_covariates = ifelse(sp_obj$dropped =
    ))
  }
  list(
    row = tibble(
      exposure = v,
      beta = tb$estimate[1],
      conf.low = if ("conf.low" %in% names(tb)) tb$conf.low[1] else NA_real_,
      conf.high = if ("conf.high" %in% names(tb)) tb$conf.high[1] else NA_real_,
      p.value = if ("p.value" %in% names(tb)) tb$p.value[1] else NA_real_,
      N = sp_obj$n_model
    ),
    diag = tibble(exposure = v, N_model = sp_obj$n_model, dropped_covariates = ifelse(sp_obj$dropped ==
  )
})
single_pollutant <- bind_rows(lapply(single_pollutant_out, `[[`, "row")) %>%
  mutate(
```

```
    beta = fmt(beta),
    conf.low = fmt(conf.low),
    conf.high = fmt(conf.high),
    p.value = fmt_p(p.value)
  )
single_pollutant_diagnostics <- bind_rows(lapply(single_pollutant_out, `[[`, "diag"))

kable(
  single_pollutant,
  caption = "Supplementary Table. Single-pollutant associations with PhenoAge acceleration (survey-weigl
)
```

Table 32: Supplementary Table. Single-pollutant associations with PhenoAge acceleration (survey-weighted, same 6-variable adjustment as Model B).

exposure	beta	conf.low	conf.high	p.value	N
LBXPFOS_log2	-0.925	-1.312	-0.538	<0.001	3058
LBXPFOA_log2	-1.399	-1.760	-1.039	<0.001	3058
LBXPFNA_log2	-0.626	-0.947	-0.305	<0.001	3058
LBXPFHS_log2	-0.809	-1.049	-0.568	<0.001	3058

```
kable(single_pollutant_diagnostics, caption = "Supplementary model diagnostics (single-pollutant droppe
```

Table 33: Supplementary model diagnostics (single-pollutant dropped covariates and fit errors).

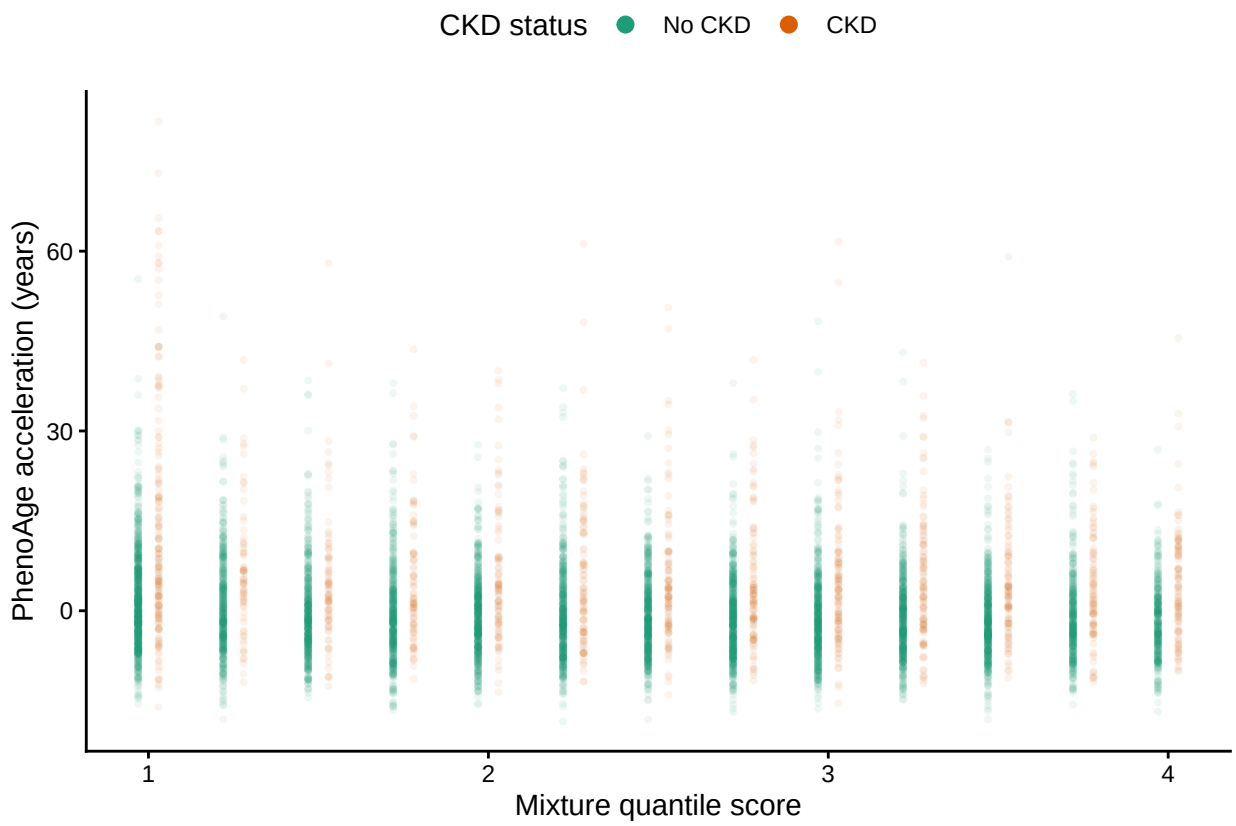
exposure	N_model	dropped_covariates	fit_error
LBXPFOS_log2	3058	None	None
LBXPFOA_log2	3058	None	None
LBXPFNA_log2	3058	None	None
LBXPFHS_log2	3058	None	None

Figures

```
dir.create("figures", showWarnings = FALSE)
n_chems <- length(primary_exposure_vars)

# Figure 1: adjusted survey-weighted predicted lines by CKD status.
pred_lines <- tryCatch({
  pdat_list <- lapply(c(0L, 1L), function(ck) {
    sub_df <- qdat %>% filter(ckd == ck)
    svy_sub <- svydesign(id = ~psu, strata = ~strata, weights = ~weight_combined, nest = TRUE, data = su
    fit_pred <- svyglm(phenoage_accel ~ mixture_qscore, design = svy_sub, family = gaussian())
    score_seq <- seq(min(sub_df$mixture_qscore, na.rm = TRUE),
                     max(sub_df$mixture_qscore, na.rm = TRUE), length.out = 100)
    newdat <- data.frame(mixture_qscore = score_seq)
```

```r
    preds <- predict(fit_pred, newdata = newdat, se.fit = TRUE, type = "response")
    newdat$fit <- as.numeric(preds$fit)
    newdat$se <- as.numeric(preds$se.fit)
    newdat$ckd <- ifelse(ck == 1L, "CKD", "No CKD")
    newdat
  })
  bind_rows(pdat_list)
}, error = function(e) NULL)

# Overall marginal direction: as mixture increases, what happens to phenoage_accel (pooled model).
overall_direction <- tryCatch({
  svy_full <- svydesign(id = ~psu, strata = ~strata, weights = ~weight_combined, nest = TRUE, data = qda
  fit_overall <- svyglm(phenoage_accel ~ mixture_qscore, design = svy_full, family = gaussian())
  score_seq <- seq(min(qdat$mixture_qscore, na.rm = TRUE), max(qdat$mixture_qscore, na.rm = TRUE), leng
  newdat <- data.frame(mixture_qscore = score_seq)
  preds <- predict(fit_overall, newdata = newdat, se.fit = TRUE, type = "response")
  newdat$fit <- as.numeric(preds$fit)
  newdat
}, error = function(e) NULL)

# Horizontal offset so green (No CKD) and orange (CKD) dots don't overlap at each quantile
x_range <- diff(range(qdat$mixture_qscore, na.rm = TRUE))
pt_offset <- max(0.005, x_range * 0.01)
qdat_plot <- qdat %>%
  mutate(
    x_plot = mixture_qscore + ifelse(ckd == 0L, -pt_offset, pt_offset),
    ckd_label = factor(ifelse(ckd == 0L, "No CKD", "CKD"), levels = c("No CKD", "CKD"))
  )

fig1 <- ggplot(qdat_plot, aes(x = x_plot, y = phenoage_accel)) +
  geom_point(aes(color = ckd_label), alpha = 0.08, size = 0.7) +
  labs(
    x = "Mixture quantile score",
    y = "PhenoAge acceleration (years)",
    color = "CKD status"
  ) +
  scale_color_manual(values = c("No CKD" = "#1b9e77", "CKD" = "#d95f02")) +
  guides(color = guide_legend(override.aes = list(alpha = 1, size = 2.5))) +
  theme_classic(base_size = 11) +
  theme(legend.position = "top")

if (!is.null(pred_lines)) {
  fig1 <- fig1 +
    geom_ribbon(
      data = pred_lines,
      aes(x = mixture_qscore, ymin = fit - 1.96 * se, ymax = fit + 1.96 * se, fill = ckd),
      alpha = 0.2, inherit.aes = FALSE
    ) +
    geom_line(
      data = pred_lines,
      aes(x = mixture_qscore, y = fit, color = ckd),
      linewidth = 1.0, inherit.aes = FALSE
    ) +
```

```
    scale_fill_manual(values = c("No CKD" = "#1b9e77", "CKD" = "#d95f02")) +
    scale_color_manual(values = c("No CKD" = "#1b9e77", "CKD" = "#d95f02"))
}
# Dotted line: overall direction of association (as mixture increases, phenoage_accel changes)
if (!is.null(overall_direction)) {
  fig1 <- fig1 +
    geom_line(
      data = overall_direction,
      aes(x = mixture_qscore, y = fit),
      linetype = "dotted", color = "gray25", linewidth = 0.9, inherit.aes = FALSE
    )
}
fig1
```




```
ggsave("figures/fig1_mixture_phenoage.jpg", plot = fig1, width = 8, height = 6, dpi = 300, device = "jp

# Figure 2: forest plot of marginal mixture effects (design-based).
fig2 <- NULL
if (!is.null(fit_int)) {
  cf_names <- names(coef(fit_int))
  int_name <- cf_names[str_detect(cf_names, "mixture_qscore:ckd")]
  if (length(int_name) == 1 && "mixture_qscore" %in% cf_names) {
    c_ckd0 <- setNames(rep(0, length(cf_names)), cf_names); c_ckd0["mixture_qscore"] <- 1
    c_ckd1 <- setNames(rep(0, length(cf_names)), cf_names); c_ckd1["mixture_qscore"] <- 1; c_ckd1[int_na
    ctr0 <- svycontrast(fit_int, c_ckd0)
```

```
    ctr1 <- svycontrast(fit_int, c_ckd1)
    ddf <- fit_int$df.residual; if (is.null(ddf) || !is.finite(ddf)) ddf <- Inf

    mk_row <- function(ctr, grp) {
      est <- as.numeric(coef(ctr)); se <- as.numeric(SE(ctr))
      tibble(
        group = grp,
        estimate = est,
        conf.low = est - qt(0.975, ddf) * se,
        conf.high = est + qt(0.975, ddf) * se,
        p.value = 2 * pt(-abs(est / se), df = ddf)
      )
    }
    forest_df <- bind_rows(
      mk_row(ctr0, "No CKD"),
      mk_row(ctr1, "CKD")
    ) %>%
      mutate(group = factor(group, levels = c("No CKD", "CKD")))

    fig2 <- ggplot(forest_df, aes(x = estimate, y = group, color = group)) +
      geom_vline(xintercept = 0, linetype = "dashed", color = "gray40") +
      geom_errorbarh(aes(xmin = conf.low, xmax = conf.high), height = 0.18, linewidth = 0.8) +
      geom_point(size = 2.8) +
      geom_text(aes(label = paste0("p=", format(round(p.value, 3), nsmall = 3))),
                nudge_y = 0.2, size = 3, color = "black", show.legend = FALSE) +
      scale_color_manual(values = c("No CKD" = "#1b9e77", "CKD" = "#d95f02")) +
      labs(
        title = "Marginal mixture effects by CKD status",
        subtitle = "Design-based contrasts from interaction model",
        x = "Beta (95% CI) for mixture score",
        y = NULL
      ) +
      theme_classic(base_size = 11) +
      theme(legend.position = "none")
    fig2
    ggsave("figures/fig2_forest_marginal_effects.jpg", plot = fig2, width = 7, height = 4, dpi = 300, d
  }
}
```

Cautions and Limitations

- Cross-sectional NHANES data do not establish temporality.
- PhenoAge includes kidney-related biomarkers (creatinine, albumin), creating potential circularity with CKD definitions. Sensitivity analyses recompute PhenoAge excluding (a) creatinine only and (b) both creatinine and albumin; both are non-standard derivatives and interpreted as robustness checks.
- PhenoAge uses Levine (2018) coefficients with correct units: albumin g/L (NHANES g/dL × 10), creatinine µmol/L (mg/dL × 88.4), glucose mmol/L (mg/dL ÷ 18.018). Model B uses phenoage_accel (age-residualized) and does not adjust for age again. Mixture quantiles are computed within each NHANES cycle to avoid temporal confounding (PFAS trends 2003–2018). Survey design uses full PSU/strata when possible; if the design fails (e.g. singleton PSUs), a weights-only fallback is used and noted.
- `qgcomp` point estimates are presented descriptively (pooled sampling weights applied, but PSU/strata

not incorporated for variance estimation). Statistical inference relies on the survey-weighted regression models (`svyglm`) above. Bootstrap variance estimation incorporating replicate weights was not implemented because NHANES does not provide replicate weights for environmental subsamples across all cycles.

- LOD handling follows NHANES detection-flag conventions when indicator columns are available (`*LC` or `*L`): values flagged below LOD (indicator == 1) are replaced with value / sqrt(2). When no LOD indicator is available, values are left unchanged (no fallback NA imputation), which improves transparency but can reduce complete-case sample size.
- **Weight selection**: a weight audit table is printed and `config$weight_type = "auto"` applies the top-coverage eligible weight. For PFAS, auto-selection excludes full-sample interview/MEC weights (`WTINT2YR`, `WTMEC2YR`) and prioritizes environmental subsample weights (`WTS*2YR`), including the combined cross-cycle PFAS weight column.
- The primary PFAS mixture was defined as a fixed four-chemical set available across all retained cycles (PFOS, PFOA, PFNA, and PFHxS), with complete-case requirement on that set. PFDA was not consistently available across cycles and was therefore excluded from the primary mixture and evaluated only in supplementary analyses. Strict complete-case (qgcomp-compatible PFAS subset) modeling is reported as sensitivity. A sensitivity analysis using an available-exposure (adaptive) mixture can be added to assess robustness when using broader PFAS information.
- Exposure quantile cutpoints are defined within cycle using cycle-specific NHANES sampling weights.
- PhenoAgeAccel is computed as the residual from a survey-weighted regression of PhenoAge on chronological age (`svyglm`), ensuring the construct is population-representative. An unweighted residualization is included as sensitivity.
- CRP unit detection uses a two-part rule: (1) table-name provenance (HSCRP tables imply mg/L), and (2) a conservative distribution heuristic (median > 1.5 **and** p75 > 3). If either triggers, CRP is converted from mg/L to mg/dL (/ 10). Cycle-level diagnostic tables are printed. PhenoAge is recomputed after any conversions. Glucose is converted from mmol/L to mg/dL if median < 6.
- Extreme biomarker values are handled by plausibility filters (WBC > 50, glucose > 600, ALP > 500 set to missing). CRP > 1 mg/dL is retained in primary analyses and evaluated in sensitivity analyses excluding potential acute inflammatory states.
- Glucose is sourced from MEC biochemistry (`glu_mec`: BIOPRO/L40) for the primary analysis. The separate GLU table (fasting) is joined with renamed variables (`LBXGLU_GLU`, `LBXSGL_GLU`) to avoid duplicate columns; it is retained for diagnostics but not used in the primary model.
- Marginal effects in the interaction model (Table 3) use `svycontrast()` with design-based degrees of freedom and t-distribution p-values, maintaining full complex-survey consistency.
- Because complete-case requirements can collapse factor levels in the interaction model frame, Model B reports both intended and realized covariate sets; unsupported factor terms are dropped with explicit logging.

Reproducibility

```r
sessionInfo()
```

```
## R version 4.5.3 (2026-03-11 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=English_United States.utf8
```

```
## [2] LC_CTYPE=English_United States.utf8
## [3] LC_MONETARY=English_United States.utf8
## [4] LC_NUMERIC=C
## [5] LC_TIME=English_United States.utf8
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] splines   grid      stats     graphics  grDevices utils     datasets
## [8] methods   base
##
## other attached packages:
##  [1] tibble_3.3.1   forcats_1.0.1  qgcomp_2.18.10 knitr_1.51     broom_1.0.12
##  [6] ggplot2_4.0.2  survey_4.5     survival_3.8-6 Matrix_1.7-4   stringr_1.6.0
## [11] purrr_1.2.1    tidyr_1.3.2    dplyr_1.2.1    nhanesA_1.4.1
##
## loaded via a namespace (and not attached):
##  [1] gtable_0.3.6        xfun_0.57           lattice_0.22-9
##  [4] numDeriv_2016.8-1.1 vctrs_0.7.2         tools_4.5.3
##  [7] Rdpack_2.6.6        generics_0.1.4      curl_7.0.0
## [10] sandwich_3.1-1      pkgconfig_2.0.3     arm_1.15-2
## [13] RColorBrewer_1.1-3  S7_0.2.1            rootSolve_1.8.2.4
## [16] lifecycle_1.0.5     compiler_4.5.3      farver_2.1.2
## [19] textshaping_1.0.5   mitools_2.4         carData_3.0-6
## [22] htmltools_0.5.9     yaml_2.3.12         Formula_1.2-5
## [25] car_3.1-5           pillar_1.11.1       nloptr_2.2.1
## [28] MASS_7.3-65         reformulas_0.4.4    boot_1.3-32
## [31] abind_1.4-8         nlme_3.1-168        tidyselect_1.2.1
## [34] rvest_1.0.5         digest_0.6.39       stringi_1.8.7
## [37] AER_1.2-16          labeling_0.4.3      fastmap_1.2.0
## [40] cli_3.6.6           magrittr_2.0.5      withr_3.0.2
## [43] foreign_0.8-91      scales_1.4.0        backports_1.5.1
## [46] rmarkdown_2.31      httr_1.4.8          otel_0.2.0
## [49] nnet_7.3-20         lme4_2.0-1          gridExtra_2.3
## [52] ragg_1.5.2          zoo_1.8-15          coda_0.19-4.1
## [55] evaluate_1.0.5      rbibutils_2.4.1     lmtest_0.9-40
## [58] pscl_1.5.9          rlang_1.2.0         Rcpp_1.1.1
## [61] glue_1.8.0          DBI_1.3.0           selectr_0.5-1
## [64] xml2_1.5.2          minqa_1.2.8         R6_2.6.1
## [67] plyr_1.8.9          systemfonts_1.3.2
```